Lab 03 – Solution Document

Working with Git History, Stashing, and Reverting Commits

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Submission Repository: https://github.com/amnajaved-dev1/DevOps

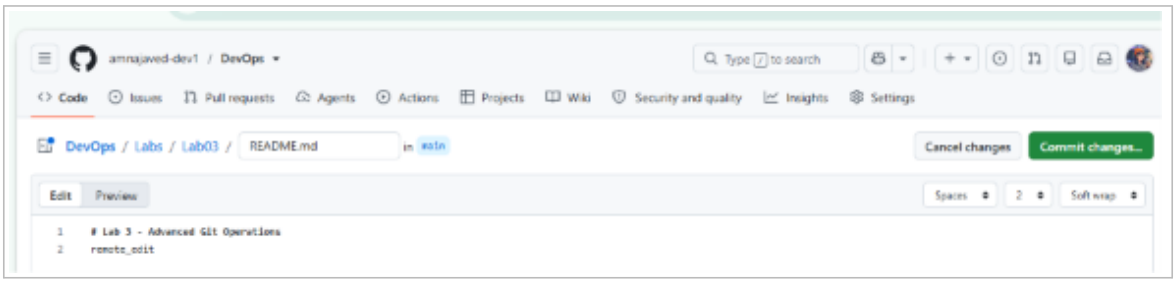
Submission Directory: Labs/Lab03/

Task 1: Handling Local and Remote Commit Conflicts (Pull vs Pull --rebase)

Objective: Understand how divergent history is handled using merge vs rebase strategies.

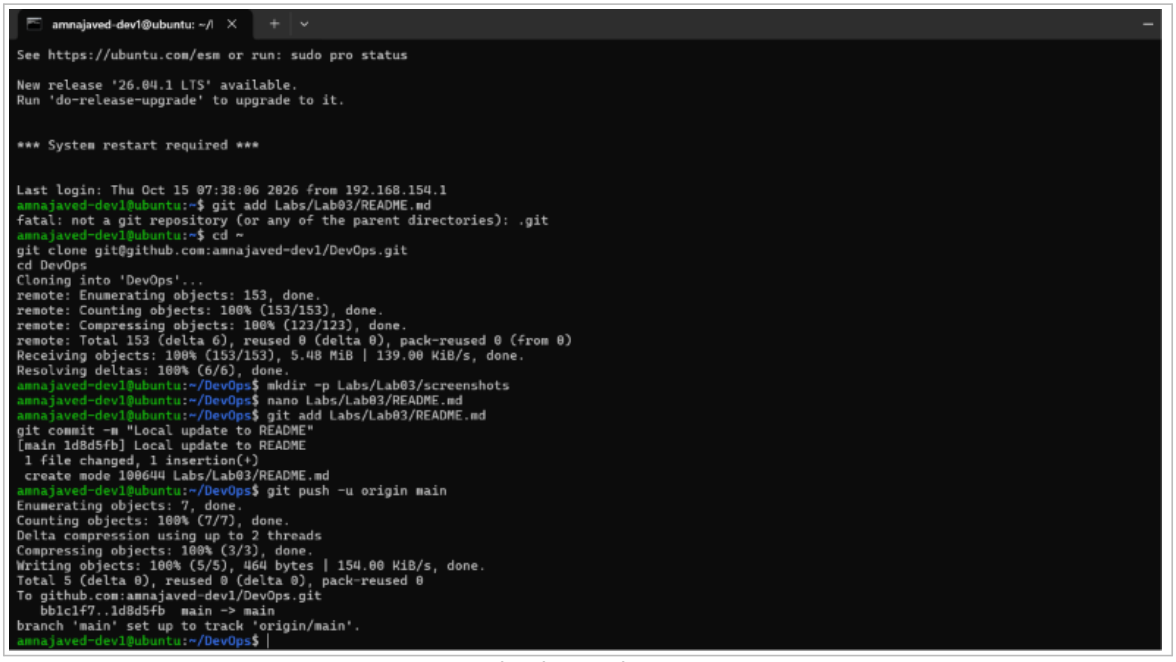
Steps Performed:

1. Edited `README.md` directly on GitHub and committed the remote change.



remote_edit.png

2. Edited the same file locally and committed the change.



local_commit.png

3. Attempted to push and observed the non-fast-forward rejection error.

```
amnajaved-dev1@ubuntu:~/DevOps$ git push origin main
To github.com:amnajaved-dev1/DevOps.git
 ! [rejected]        main -> main (fetch first)
error: failed to push some refs to 'github.com:amnajaved-dev1/DevOps.git'
hint: Updates were rejected because the remote contains work that you do not
hint: have locally. This is usually caused by another repository pushing to
hint: the same ref. If you want to integrate the remote changes, use
hint: 'git pull' before pushing again.
hint: See the 'Note about fast-forwards' in 'git push --help' for details.
amnajaved-dev1@ubuntu:~/DevOps$
```

push_error.png

4. Pulled changes using standard merge (`git pull --no-rebase origin main`) and successfully pushed.

```
amnajaved-dev1@ubuntu:~/DevOps$ git pull --no-rebase origin main
remote: Enumerating objects: 9, done.
remote: Counting objects: 100% (9/9), done.
remote: Compressing objects: 100% (3/3), done.
remote: Total 5 (delta 1), reused 0 (delta 0), pack-reused 0 (from 0)
Unpacking objects: 100% (5/5), 1.06 KiB | 120.00 KiB/s, done.
From github.com:amnajaved-dev1/DevOps
 * branch            main       -> FETCH_HEAD
   1d8d5fb..015e935  main       -> origin/main
Updating 1d8d5fb..015e935
Fast-forward
 Labs/Lab03/README.md | 1 +
 1 file changed, 1 insertion(+)
```

merge_commit.png

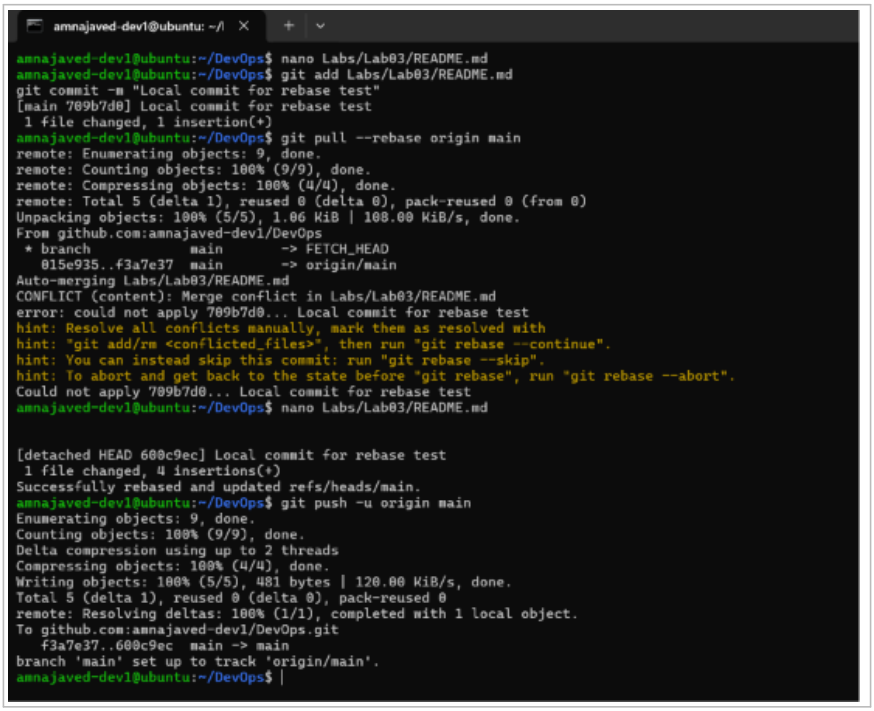
```
amnajaved-dev1@ubuntu:~/DevOps$ git push -u origin main
branch 'main' set up to track 'origin/main'.
Everything up-to-date
amnajaved-dev1@ubuntu:~/DevOps$
```

push_after_merge.png

5. Tested the rebase workflow (`git pull --rebase origin main`) to keep a linear commit history and pushed.

```
amnajaved-dev1@ubuntu:~/DevOps$ git push -u origin main
branch 'main' set up to track 'origin/main'.
Everything up-to-date
amnajaved-dev1@ubuntu:~/DevOps$ nano Labs/Lab03/README.md
amnajaved-dev1@ubuntu:~/DevOps$ git add Labs/Lab03/README.md
git commit -m "Local commit for rebase test"
[main 709b7d0] Local commit for rebase test
 1 file changed, 1 insertion(+)
amnajaved-dev1@ubuntu:~/DevOps$ git pull --rebase origin main
remote: Enumerating objects: 9, done.
remote: Counting objects: 100% (9/9), done.
remote: Compressing objects: 100% (4/4), done.
remote: Total 5 (delta 1), reused 0 (delta 0), pack-reused 0 (from 0)
Unpacking objects: 100% (5/5), 1.06 KiB | 108.00 KiB/s, done.
From github.com:amnajaved-dev1/DevOps
 * branch            main       -> FETCH_HEAD
   015e935..f3a7e37  main       -> origin/main
Auto-merging Labs/Lab03/README.md
CONFLICT (content): Merge conflict in Labs/Lab03/README.md
error: could not apply 709b7d0... Local commit for rebase test
hint: Resolve all conflicts manually, mark them as resolved with
hint: "git add/rm <conflicted_files>", then run "git rebase --continue".
hint: You can instead skip this commit: run "git rebase --skip".
hint: To abort and get back to the state before "git rebase", run "git rebase --abort".
Could not apply 709b7d0... Local commit for rebase test
amnajaved-dev1@ubuntu:~/DevOps$
```

rebase_pull.png



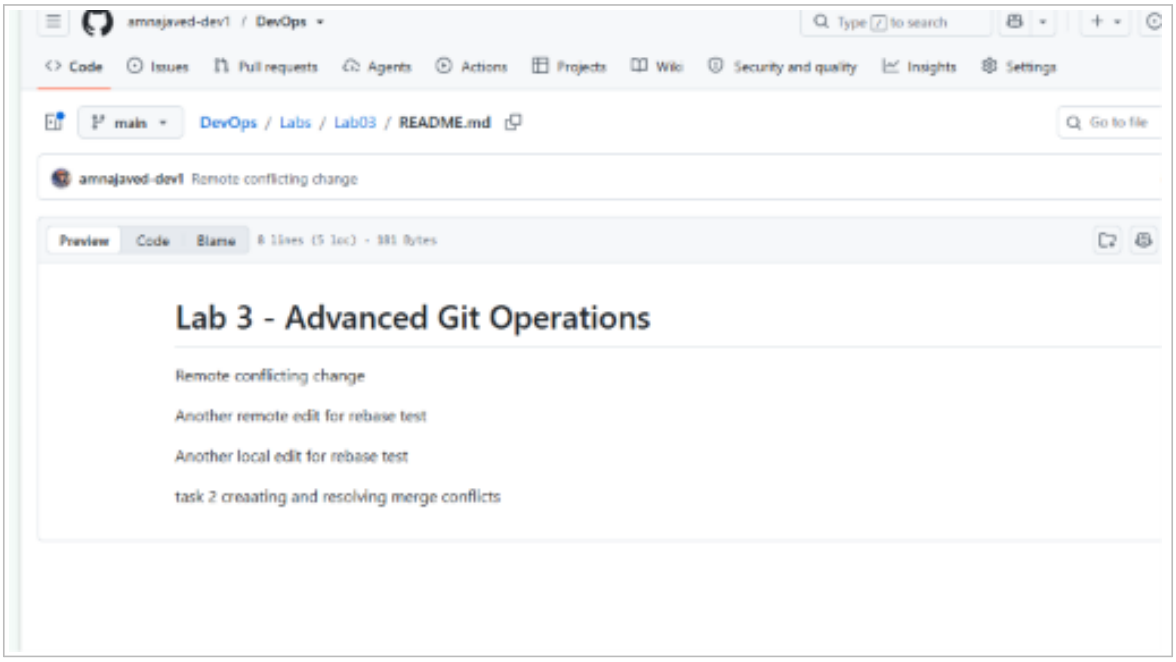
push_after_rebase.png

Task 2: Creating and Resolving Merge Conflicts Manually

Objective: Resolve overlapping edits on the same file line manually.

Steps Performed:

1. Made a conflicting edit on GitHub.



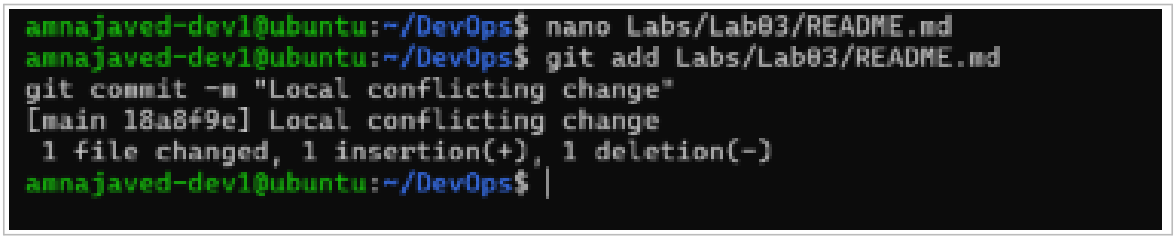
remote_conflict_edit.png

2. Made a conflicting edit locally on the exact same line.

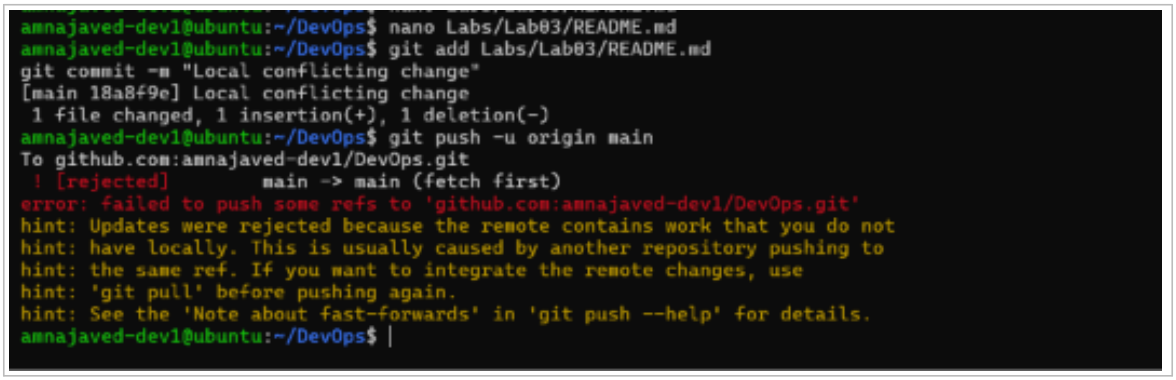
[Screenshot not found: local_conflict_edit.png]

local_conflict_edit.png

3. Committed the local change and attempted to push (resulting in rejection).

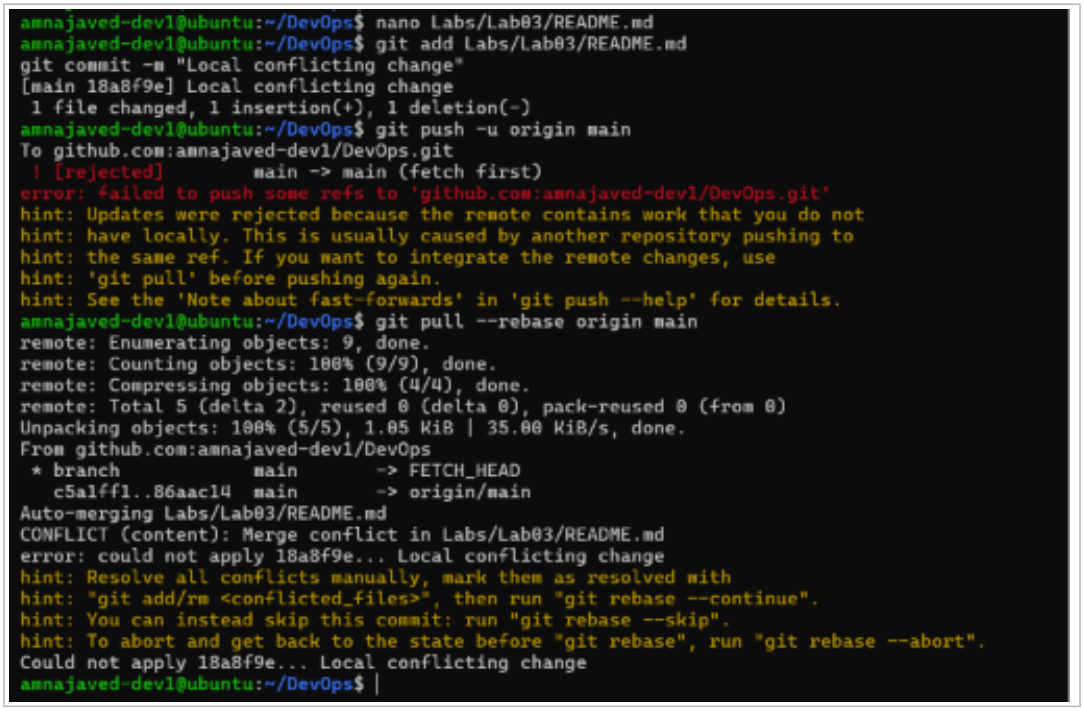


local_conflict_commit.png



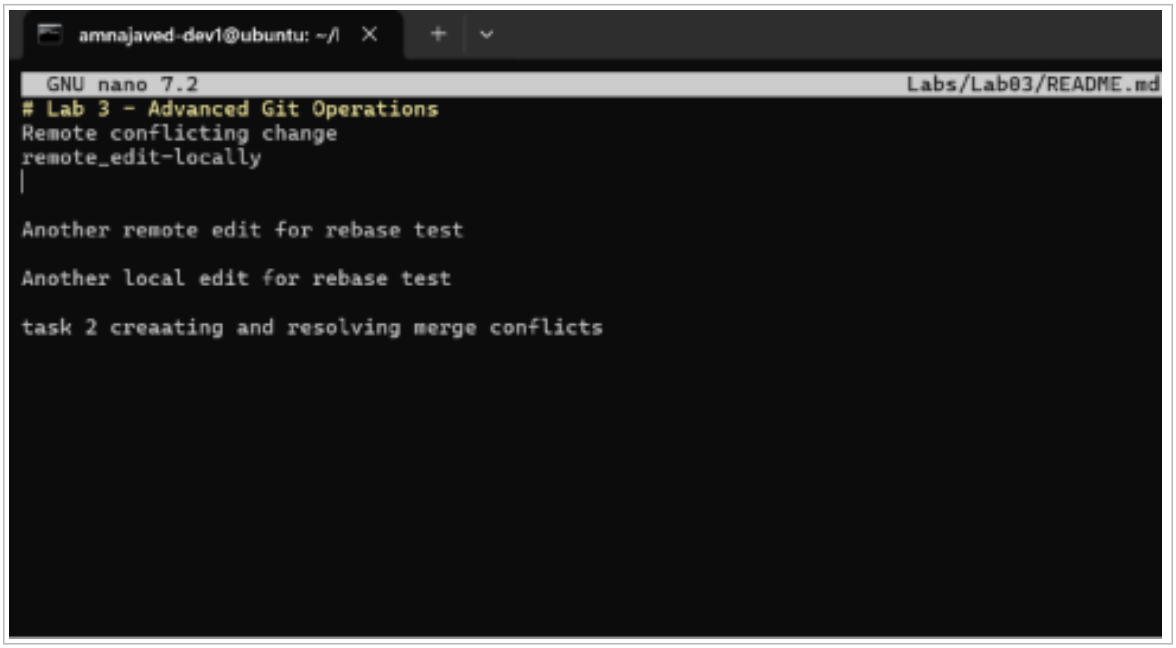
conflict_push_error.png

4. Ran `git pull --rebase origin main` to trigger the conflict.



conflict_message.png

5. Manually resolved the conflict markers (<<<<<<<, =======, >>>>>>>) in VS Code.

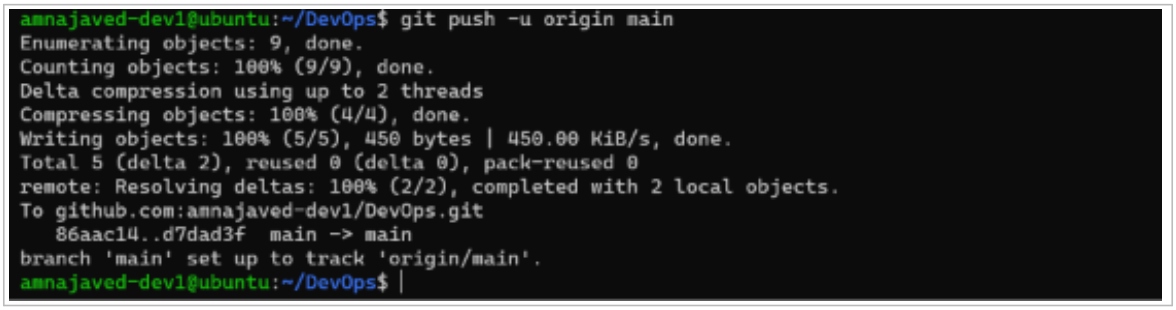


resolved_readme.png

6. Continued the rebase (`git add` and `git rebase --continue`) and pushed to remote.



rebase_continue.png



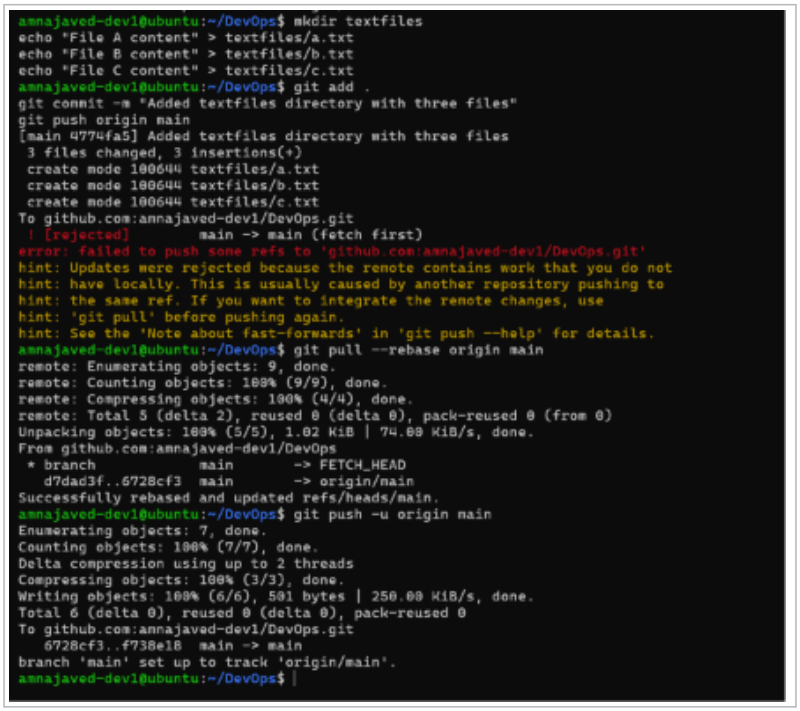
push_after_resolve.png

Task 3: Managing Ignored Files with .gitignore and Removing Tracked Files

Objective: Exclude specific directories using `.gitignore` and untrack previously committed files.

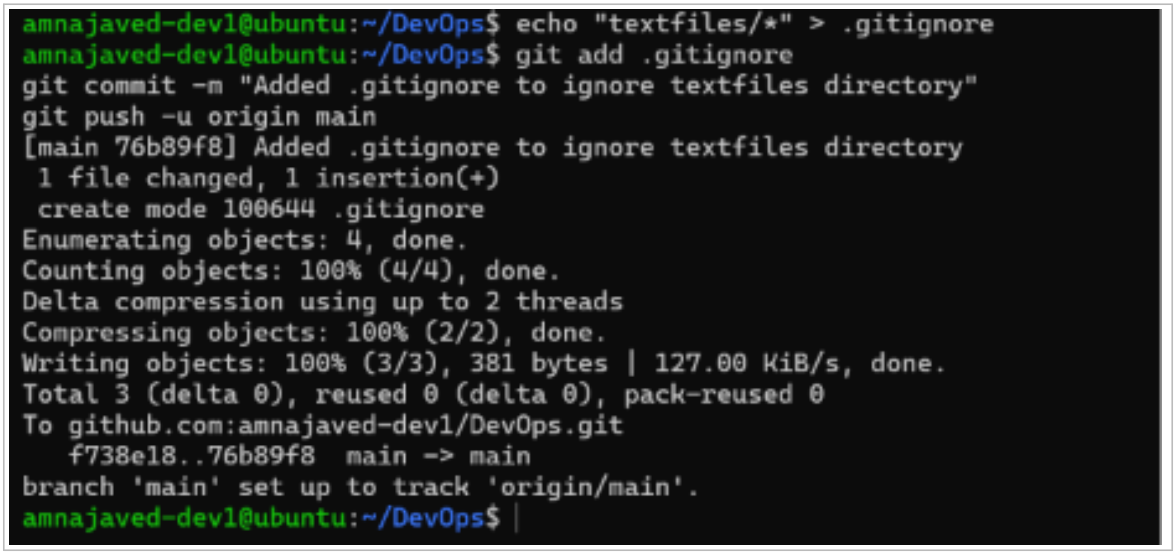
Steps Performed:

1. Created a `textfiles/` directory with sample `.txt` files and pushed them.



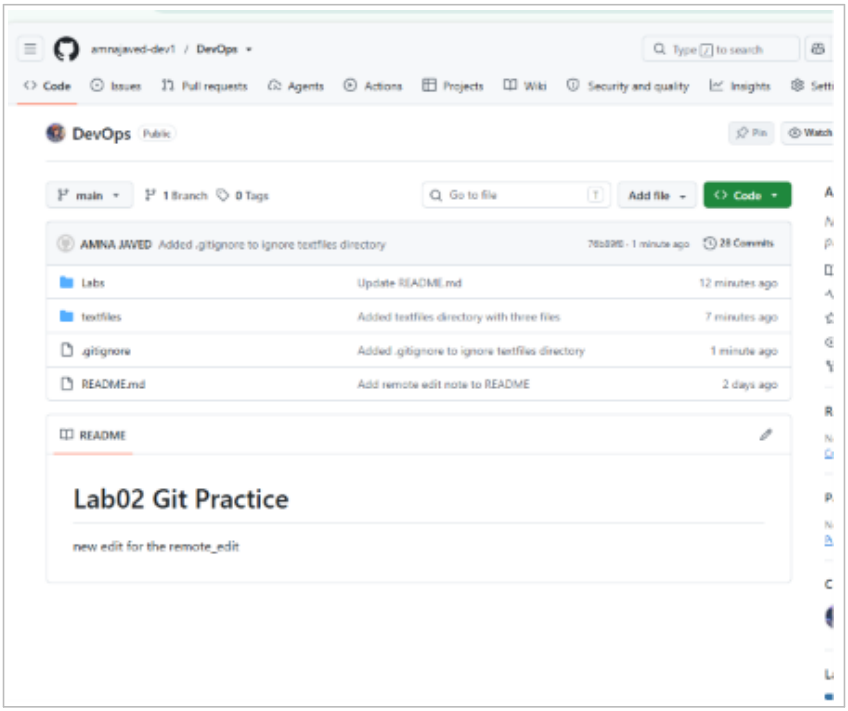
push_textfiles.png

2. Created a `.gitignore` file to ignore `textfiles/*` and pushed the update.



gitignore_push.png

3. Verified that the folder remained visible on GitHub despite `.gitignore`.



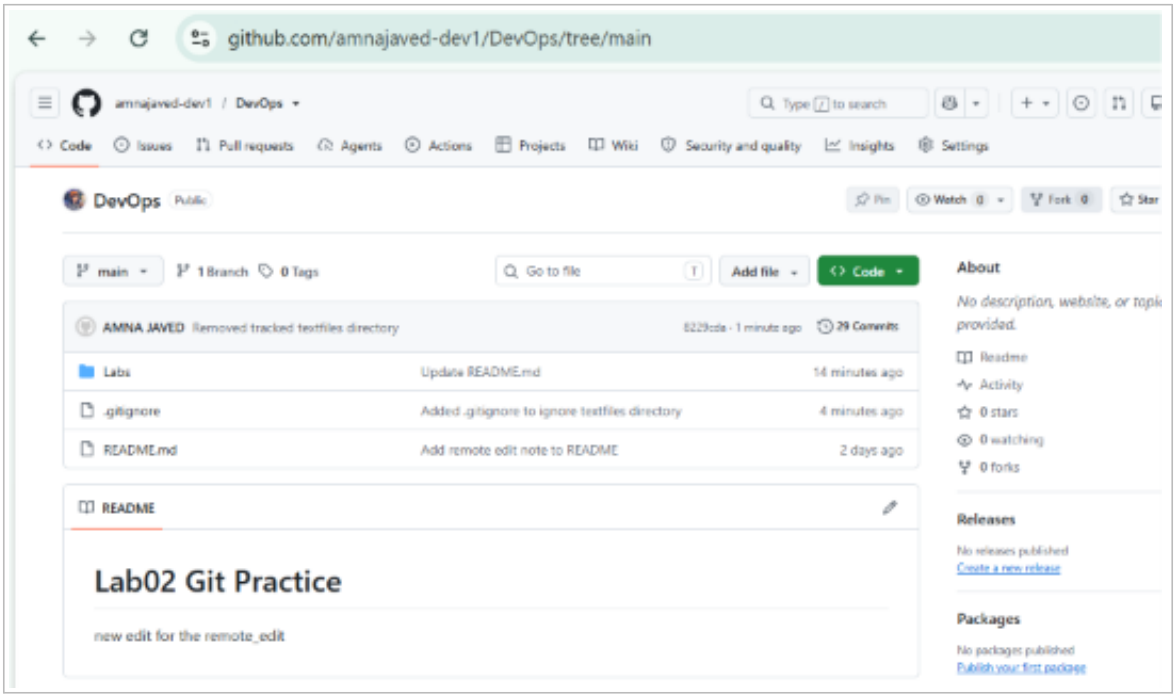
repo_still_has_textfiles.png

4. Untracked the folder from Git cache using `git rm -r --cached textfiles` while retaining local files.

```
amnajaved-dev1@ubuntu:~/DevOps$ git rm -r --cached textfiles
rm 'textfiles/a.txt'
rm 'textfiles/b.txt'
rm 'textfiles/c.txt'
amnajaved-dev1@ubuntu:~/DevOps$ git add .
git commit -m "Removed tracked textfiles directory"
git push -u origin main
[main 8229cda] Removed tracked textfiles directory
 3 files changed, 3 deletions(-)
 delete mode 100644 textfiles/a.txt
 delete mode 100644 textfiles/b.txt
 delete mode 100644 textfiles/c.txt
Enumerating objects: 3, done.
Counting objects: 100% (3/3), done.
Delta compression using up to 2 threads
Compressing objects: 100% (2/2), done.
Writing objects: 100% (2/2), 246 bytes | 246.00 KiB/s, done.
Total 2 (delta 1), reused 0 (delta 0), pack-reused 0
remote: Resolving deltas: 100% (1/1), completed with 1 local object.
To github.com:amnajaved-dev1/DevOps.git
   76b89f8..8229cda  main -> main
branch 'main' set up to track 'origin/main'.
amnajaved-dev1@ubuntu:~/DevOps$
```

rm_cached_push.png

5. Verified successful removal from the remote repository.



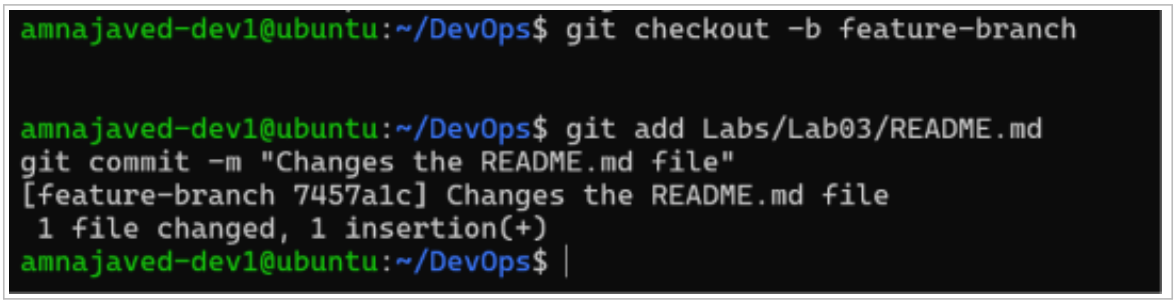
repo_textfiles_removed.png

Task 4: Create Temporary Changes and Use git stash

Objective: Safely store uncommitted work to switch branches without data loss.

Steps Performed:

1. Modified `README.md` on a new feature branch.



modified_readme.png

2. Attempted to switch to `main` without committing/stashing, triggering a block error.

```
amnajaved-dev1@ubuntu:~/DevOps$ git checkout -b feature-branch


amnajaved-dev1@ubuntu:~/DevOps$ git add Labs/Lab03/README.md
git commit -m "Changes the README.md file"
[feature-branch 7457a1c] Changes the README.md file


amnajaved-dev1@ubuntu:~/DevOps$ git checkout main
error: Your local changes to the following files would be overwritten by checkout:
        Labs/Lab03/README.md
Please commit your changes or stash them before you switch branches.
Aborting
amnajaved-dev1@ubuntu:~/DevOps$ nano Labs/Lab03/README.md
amnajaved-dev1@ubuntu:~/DevOps$ git checkout main
error: Your local changes to the following files would be overwritten by checkout:
        Labs/Lab03/README.md
Please commit your changes or stash them before you switch branches.
Aborting
amnajaved-dev1@ubuntu:~/DevOps$
```

checkout_error.png

3. Stashed changes using `git stash`.

```
amnajaved-dev1@ubuntu:~/DevOps$ git stash
Saved working directory and index state WIP on feature-branch: 7457a1c Changes the README.md file
amnajaved-dev1@ubuntu:~/DevOps$
```

stash_command.png

4. Switched successfully to `main` and returned to `feature-branch`.

```
amnajaved-dev1@ubuntu:~/DevOps$ git checkout main
Switched to branch 'main'
Your branch is up to date with 'origin/main'.
amnajaved-dev1@ubuntu:~/DevOps$
```

branch_switched.png

```
amnajaved-dev1@ubuntu:~/DevOps$ git stash
Saved working directory and index state WIP on feature-branch: 7457a1c Changes the README.md file
amnajaved-dev1@ubuntu:~/DevOps$ git checkout main
Switched to branch 'main'
Your branch is up to date with 'origin/main'.
amnajaved-dev1@ubuntu:~/DevOps$ git checkout feature-branch
Switched to branch 'feature-branch'
amnajaved-dev1@ubuntu:~/DevOps$
```

back_to_feature.png

```
amnajaved-dev1@ubuntu:~/DevOps$ git stash
Saved working directory and index state WIP on feature-branch: 7457a1c Changes the README.md file
amnajaved-dev1@ubuntu:~/DevOps$ git checkout main
Switched to branch 'main'
Your branch is up to date with 'origin/main'.
amnajaved-dev1@ubuntu:~/DevOps$ git checkout feature-branch
Switched to branch 'feature-branch'
amnajaved-dev1@ubuntu:~/DevOps$ git status
On branch feature-branch
nothing to commit, working tree clean
amnajaved-dev1@ubuntu:~/DevOps$
```

status_clean.png

5. Restored stashed changes using `git stash pop`.

```
amnajaved-dev1@ubuntu:~/DevOps$ git status
On branch feature-branch
nothing to commit, working tree clean
amnajaved-dev1@ubuntu:~/DevOps$ git stash pop
On branch feature-branch
Changes not staged for commit:
  (use "git add <file>..." to update what will be committed)
  (use "git restore <file>..." to discard changes in working directory)
        modified:   Labs/Lab03/README.md

no changes added to commit (use "git add" and/or "git commit -a")
Dropped refs/stash@{0} (565dd7ce009ca945adab6618a840f04c941bd206)
amnajaved-dev1@ubuntu:~/DevOps$
```

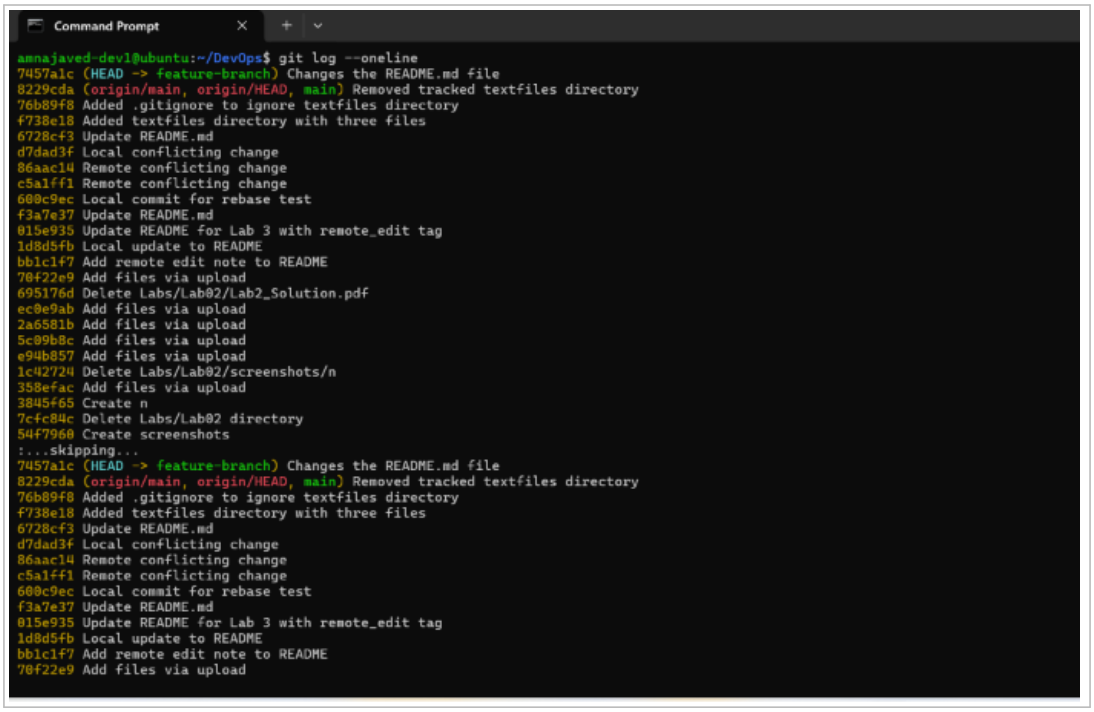
stash_pop.png

Task 5: Checkout a Specific Commit Using git log

Objective: Inspect historical commits in a detached HEAD state.

Steps Performed:

1. Viewed commit logs using `git log --oneline`.



log_before_checkout.png

2. Checked out a specific historical commit hash.

```
amnajaved-dev1@ubuntu:~/DevOps$ git stash
git checkout f738e18
Saved working directory and index state WIP on feature-branch: 7457a1c Changes the README.md file
Note: switching to 'f738e18'.

You are in 'detached HEAD' state. You can look around, make experimental
changes and commit them, and you can discard any commits you make in this
state without impacting any branches by switching back to a branch.

If you want to create a new branch to retain commits you create, you may
do so (now or later) by using -c with the switch command. Example:

  git switch -c <new-branch-name>

Or undo this operation with:

  git switch -

Turn off this advice by setting config variable advice.detachedHead to false

HEAD is now at f738e18 Added textfiles directory with three files
amnajaved-dev1@ubuntu:~/DevOps$
```

detached_head.png

3. Returned safely to the `main` branch.

```
amnajaved-dev1@ubuntu:~/DevOps$ git checkout main
Previous HEAD position was f738e18 Added textfiles directory with three files
Switched to branch 'main'
Your branch is up to date with 'origin/main'.
amnajaved-dev1@ubuntu:~/DevOps$
```

back_to_main.png

Task 6: Resetting Commits (Soft vs Hard Reset)

Objective: Compare soft resets (retaining working files) versus hard resets (discarding changes).

Steps Performed:

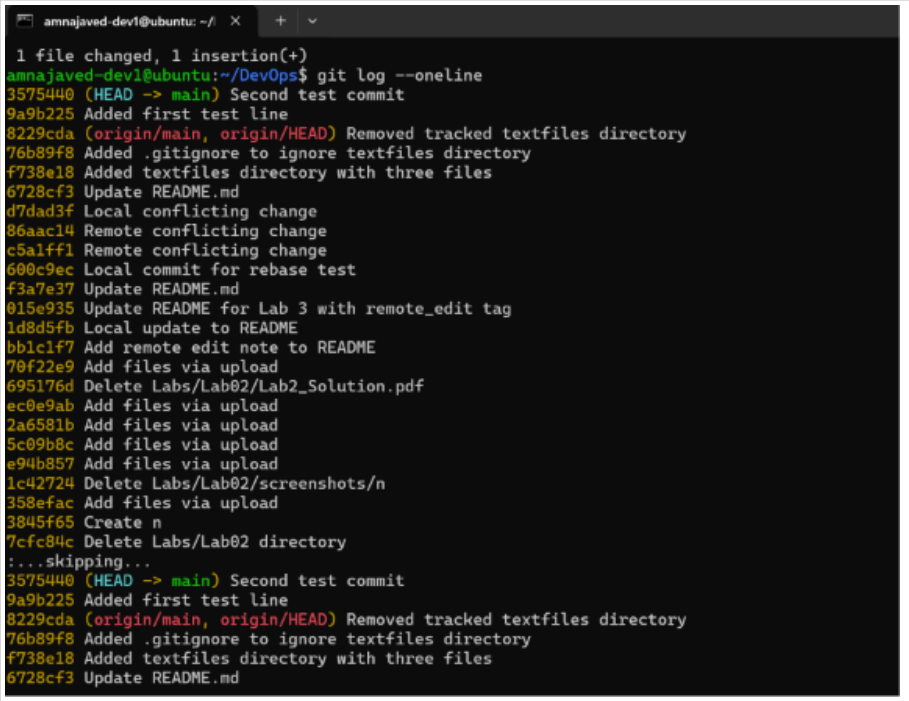
1. Made and committed two consecutive test changes.

```
amnajaved-dev1@ubuntu:~/DevOps$ echo "First test line" >> Labs/Lab03/README.md
git add .
git commit -m "Added first test line"
[main 9a9b225] Added first test line
 1 file changed, 1 insertion(+)
amnajaved-dev1@ubuntu:~/DevOps$
```

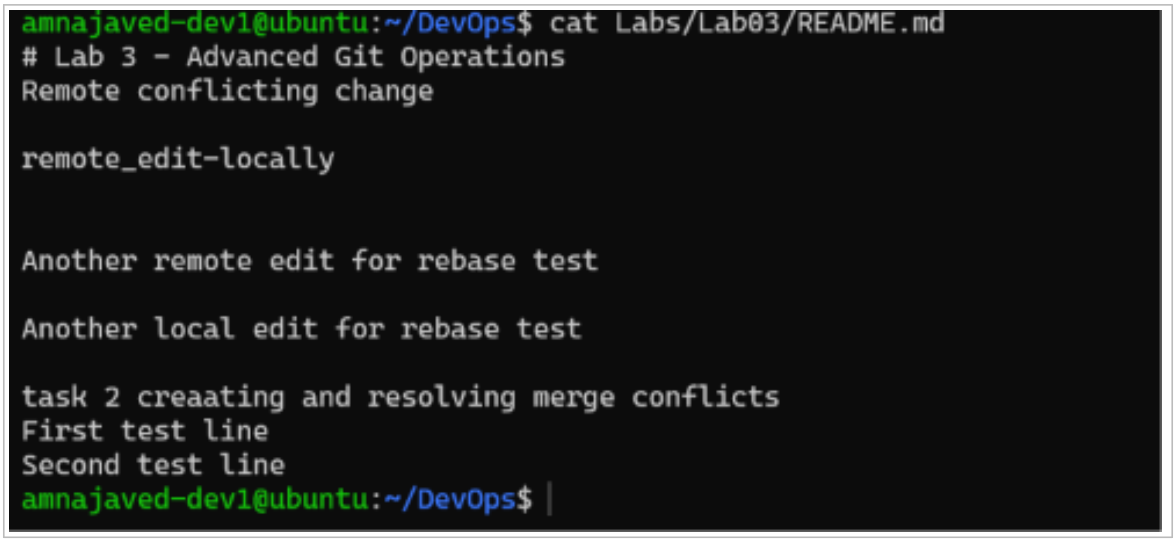
first_commit.png

```
amnajaved-dev1@ubuntu:~/DevOps$ echo "Second test line" >> Labs/Lab03/README.md
git add .
git commit -m "Second test commit"
[main 3575440] Second test commit
 1 file changed, 1 insertion(+)
amnajaved-dev1@ubuntu:~/DevOps$
```

second_commit.png



log_before_reset.png



file_before_reset.png

2. Performed a soft reset (`git reset --soft HEAD~1`), verified logs, file contents, and staged status.

```
amnajaved-dev1@ubuntu:~/DevOps$ cat Labs/Lab03/README.md
# Lab 3 - Advanced Git Operations
Remote conflicting change

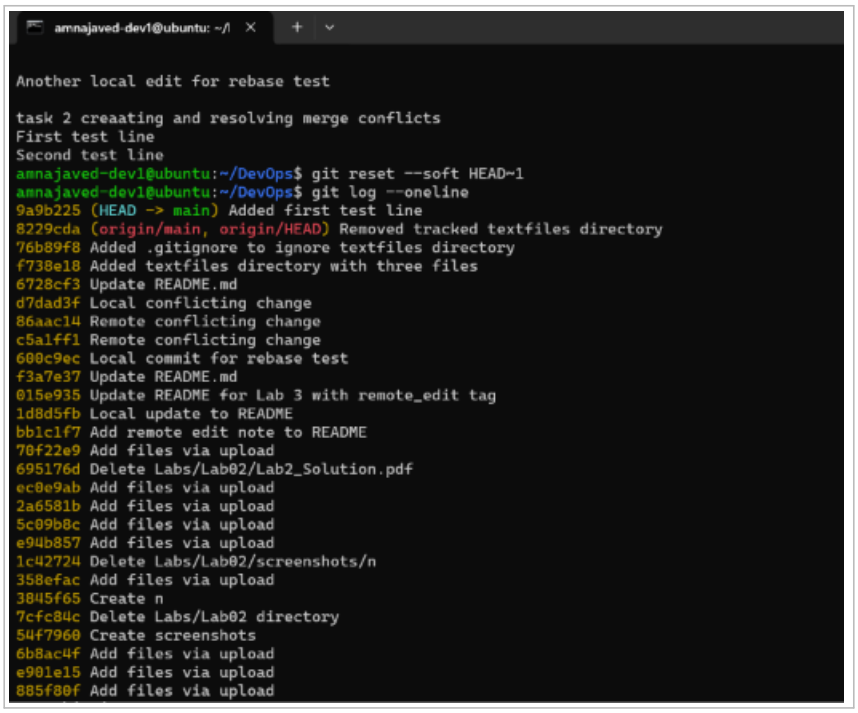
remote_edit-locally


Another remote edit for rebase test

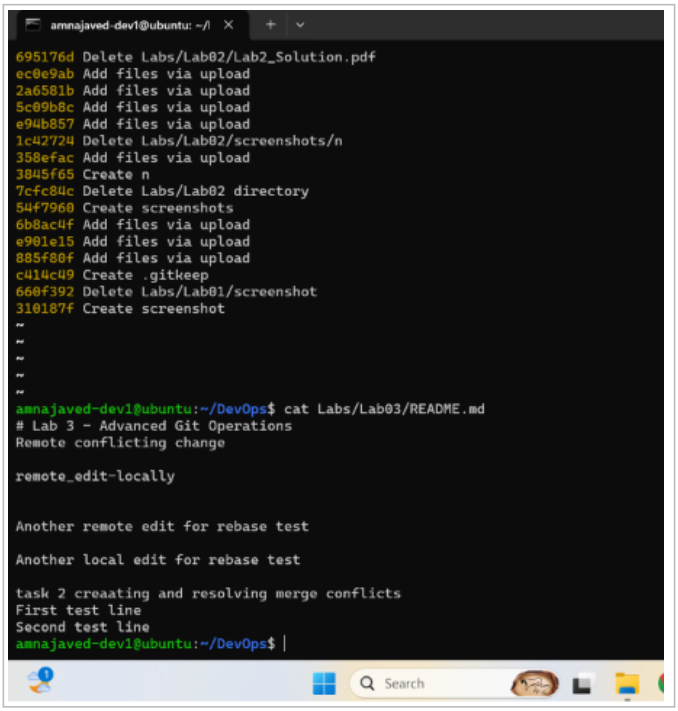
Another local edit for rebase test

task 2 creaating and resolving merge conflicts
First test line
Second test line
amnajaved-dev1@ubuntu:~/DevOps$ git reset --soft HEAD~1
amnajaved-dev1@ubuntu:~/DevOps$
```

soft_reset.png

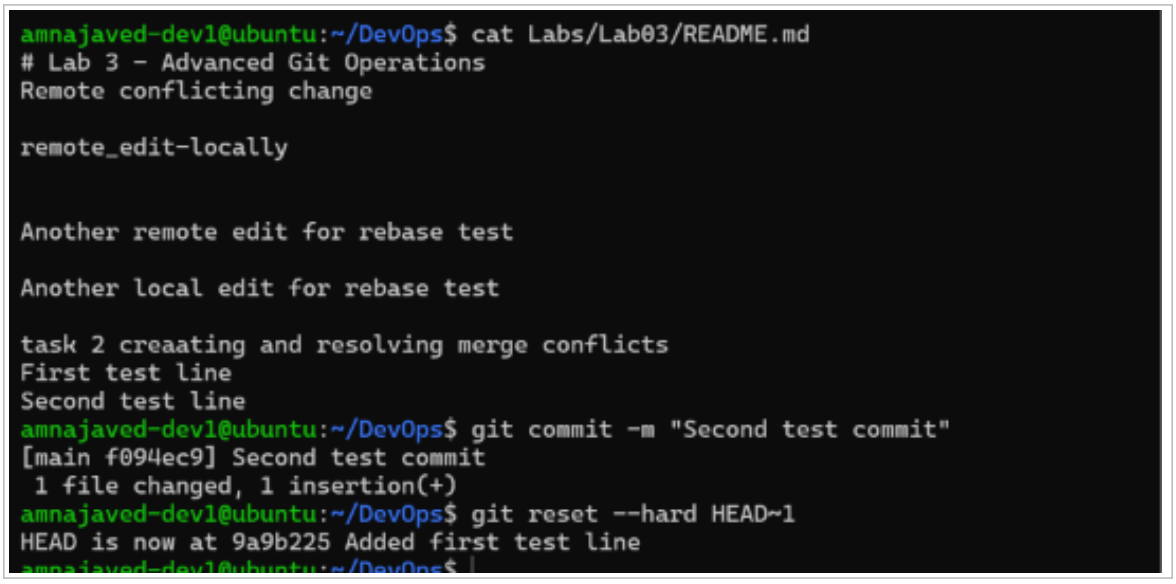


log_after_soft_reset.png

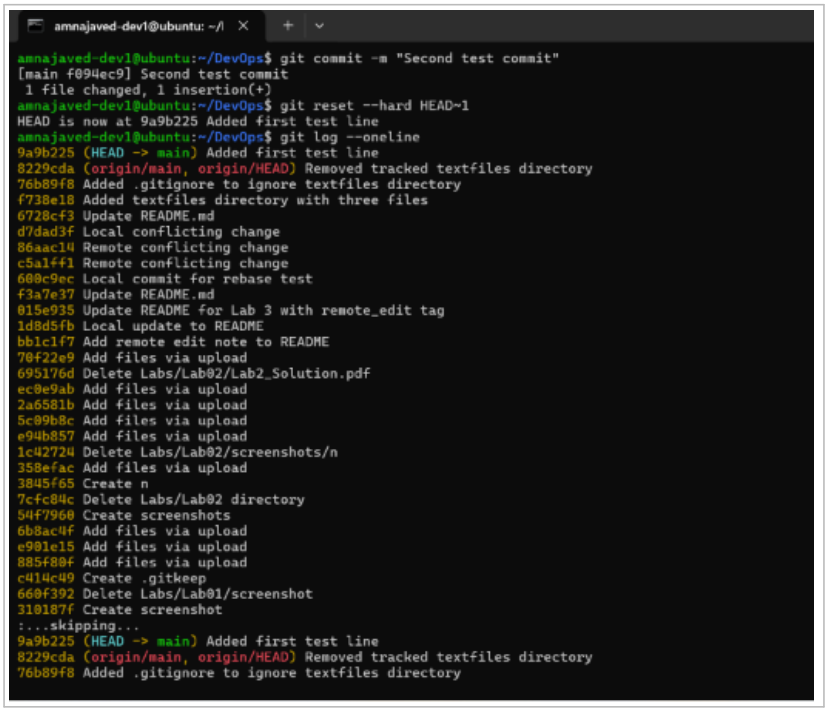


file_after_soft_reset.png

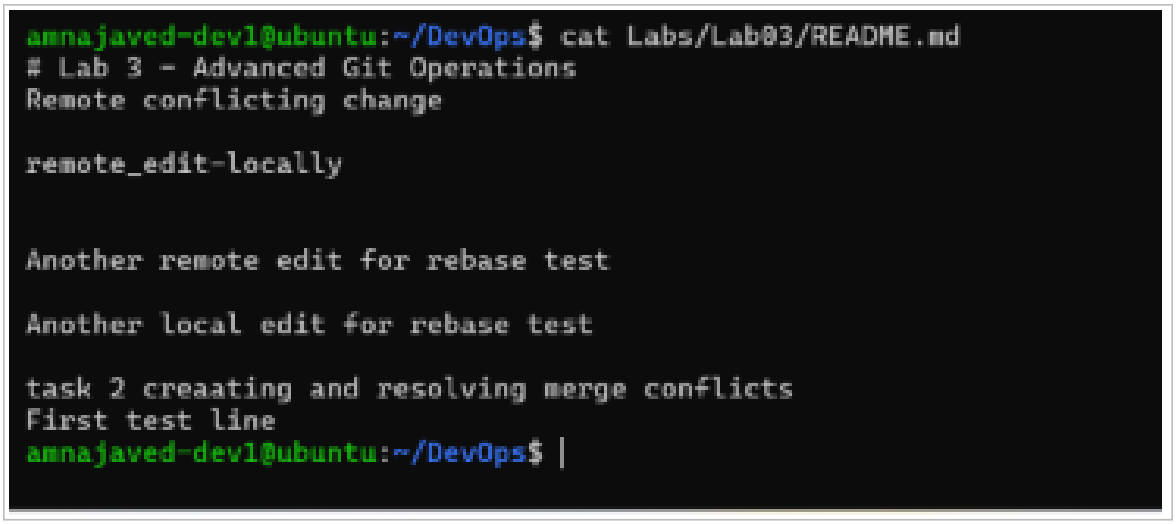
3. Recommitted and performed a hard reset (`git reset --hard HEAD~1`), verifying that changes were completely discarded.



hard_reset.png



log_after_hard_reset.png



file_after_hard_reset.png

Task 7: Amending the Last Commit

Objective: Modify the most recent commit message or include forgotten files.

Steps Performed:

1. Committed a file update with a fix message.

```
amnajaved-dev1@ubuntu:~/DevOps$ git add .
git commit -m "Fix log message"
On branch main
Your branch is ahead of 'origin/main' by 1 commit.
  (use "git push" to publish your local commits)

nothing to commit, working tree clean
amnajaved-dev1@ubuntu:~/DevOps$
```

first_amend_commit.png

2. Added additional changes and amended the commit using `git commit --amend`.

```
amnajaved-dev1@ubuntu:~/DevOps$ git add .
git commit -m "Fix log message"
On branch main
Your branch is ahead of 'origin/main' by 1 commit.
  (use "git push" to publish your local commits)

nothing to commit, working tree clean
amnajaved-dev1@ubuntu:~/DevOps$ git add .
git commit --amend -m "Updated commit with additional fix"
[main 5084fe6] Updated commit with additional fix
 Date: Sat Oct 17 13:42:48 2026 +0000
 1 file changed, 1 insertion(+)
amnajaved-dev1@ubuntu:~/DevOps$
```

amend_commit.png

Task 8: Reverting a Commit (Safe Undo on Remote Branch)

Objective: Safely roll back shared history using `git revert` without rewriting commit history.

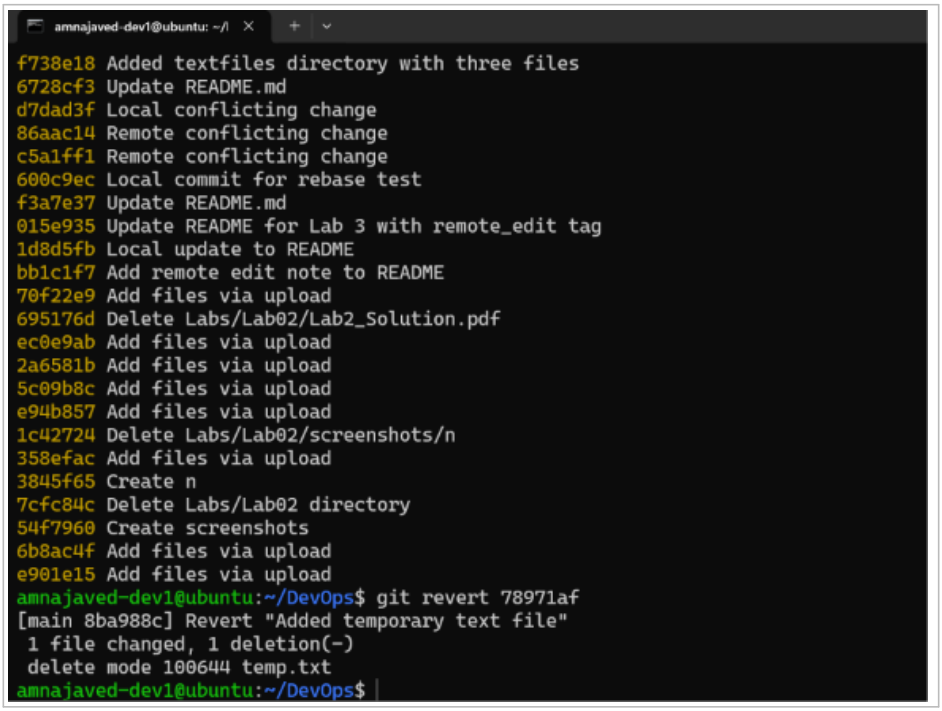
Steps Performed:

1. Created and pushed `temp.txt`.

```
amnajaved-dev1@ubuntu:~/DevOps$ echo "temporary text" >> temp.txt
git add .
git commit -m "Added temporary text file"
git push origin main
[main 78971af] Added temporary text file
 1 file changed, 1 insertion(+)
 create mode 100644 temp.txt
Enumerating objects: 12, done.
Counting objects: 100% (12/12), done.
Delta compression using up to 2 threads
Compressing objects: 100% (6/6), done.
Writing objects: 100% (8/8), 724 bytes | 38.00 KiB/s, done.
Total 8 (delta 3), reused 0 (delta 0), pack-reused 0
remote: Resolving deltas: 100% (3/3), completed with 2 local objects.
To github.com:amnajaved-dev1/DevOps.git
   8229cda..78971af  main -> main
amnajaved-dev1@ubuntu:~/DevOps$
```

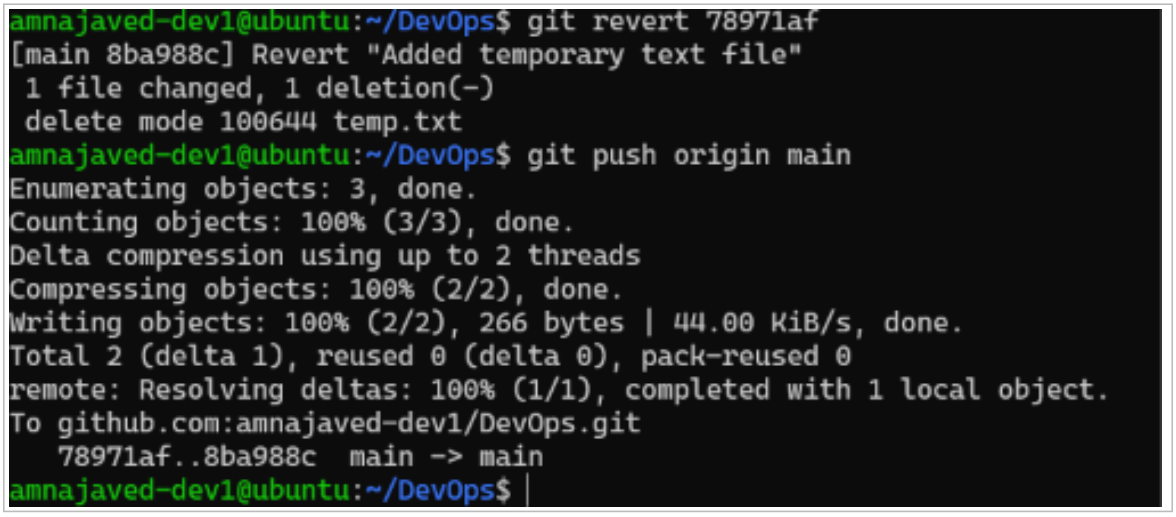
commit_temp_file.png

2. Reverted the commit using `git revert <hash>`.



revert_commit.png

3. Pushed the revert commit to remote.



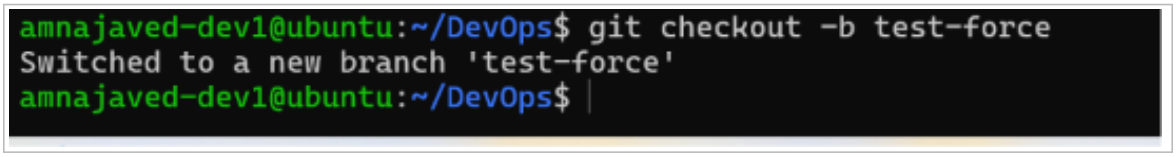
revert_push.png

Task 9: Force Push (With Caution)

Objective: Rewrite remote history safely on an isolated test branch.

Steps Performed:

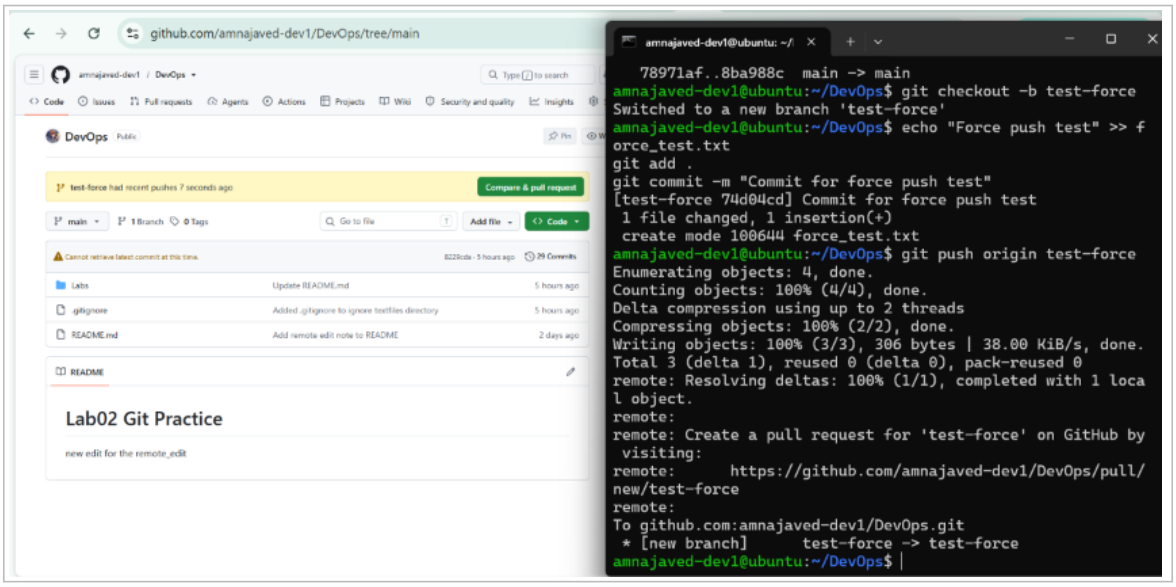
1. Created a new branch `test-force`, committed a change, and pushed.



new_branch.png

```
amnajaved-dev1@ubuntu:~/DevOps$ git checkout -b test-force
Switched to a new branch 'test-force'
amnajaved-dev1@ubuntu:~/DevOps$ echo "Force push test" >> force_test.txt
git add .
git commit -m "Commit for force push test"
[test-force 74d04cd] Commit for force push test
 1 file changed, 1 insertion(+)
 create mode 100644 force_test.txt
amnajaved-dev1@ubuntu:~/DevOps$
```

force_commit.png

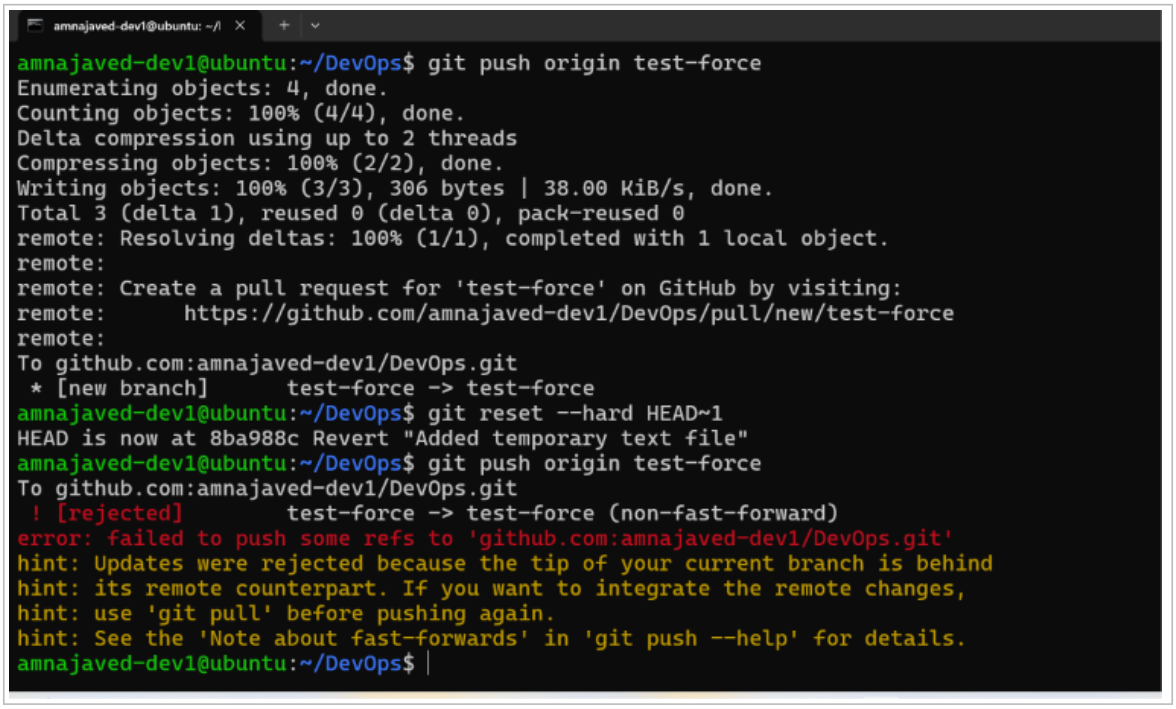


push_force_branch.png

2. Performed a hard reset locally and attempted a normal push (which was rejected).

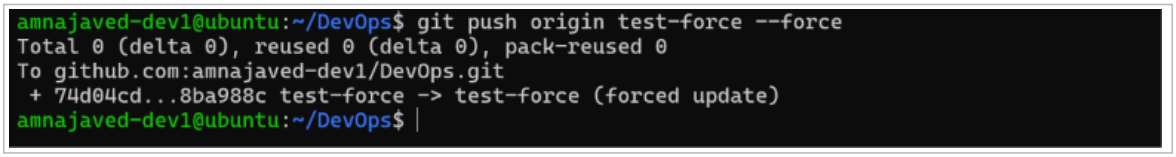
```
amnajaved-dev1@ubuntu:~/DevOps$ git push origin test-force
Enumerating objects: 4, done.
Counting objects: 100% (4/4), done.
Delta compression using up to 2 threads
Compressing objects: 100% (2/2), done.
Writing objects: 100% (3/3), 306 bytes | 38.00 KiB/s, done.
Total 3 (delta 1), reused 0 (delta 0), pack-reused 0
remote: Resolving deltas: 100% (1/1), completed with 1 local object.
remote:
remote: Create a pull request for 'test-force' on GitHub by visiting:
remote:      https://github.com/amnajaved-dev1/DevOps/pull/new/test-force
remote:
To github.com:amnajaved-dev1/DevOps.git
 * [new branch]      test-force -> test-force
amnajaved-dev1@ubuntu:~/DevOps$ git reset --hard HEAD~1
HEAD is now at 8ba988c Revert "Added temporary text file"
amnajaved-dev1@ubuntu:~/DevOps$
```

hard_reset_force.png



normal_push.png

3. Force-pushed successfully using `git push origin test-force --force`.



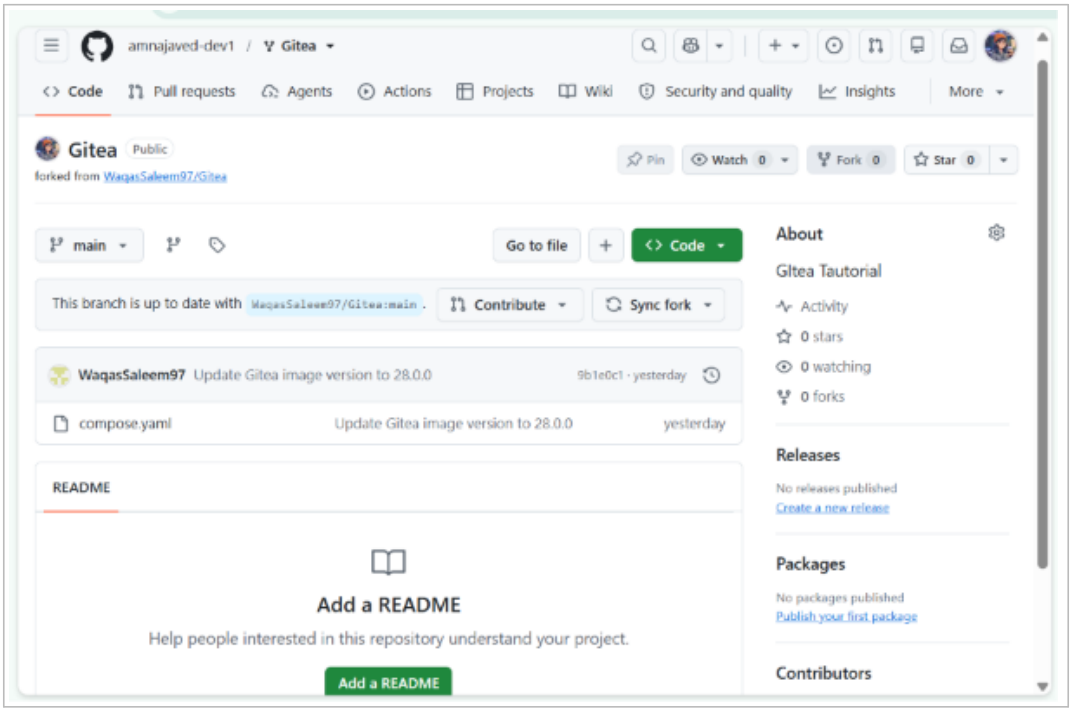
force_push.png

Task 10: Running Gitea in GitHub Codespaces via Docker Compose

Objective: Fork and run a self-hosted Git service (Gitea) inside a containerized Codespace environment.

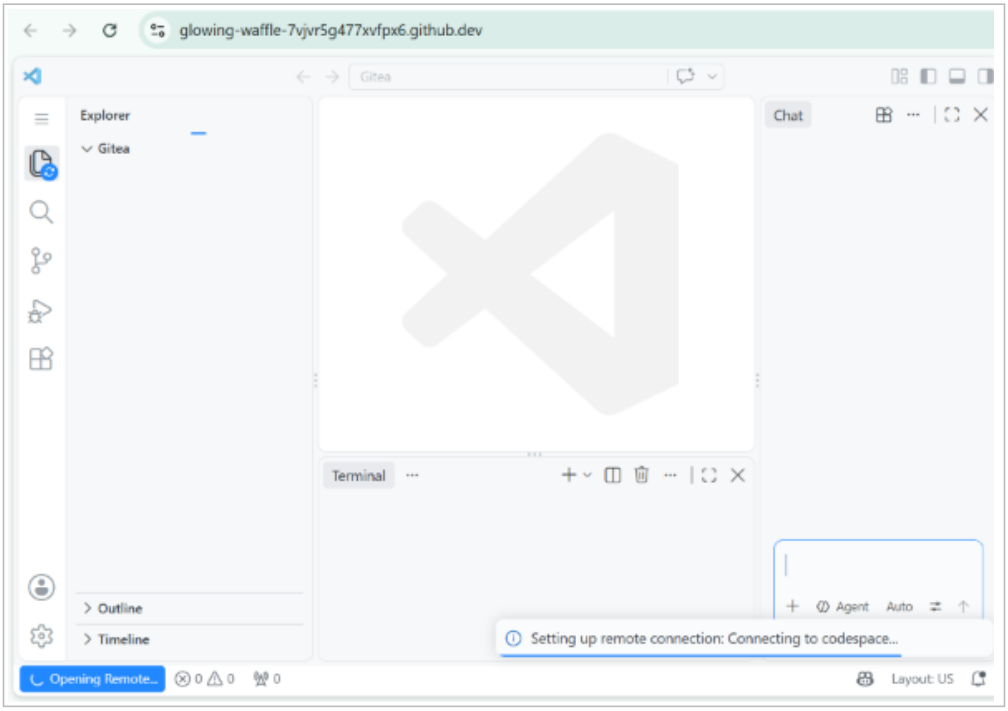
Steps Performed:

1. Forked the Gitea repository with all branches included.



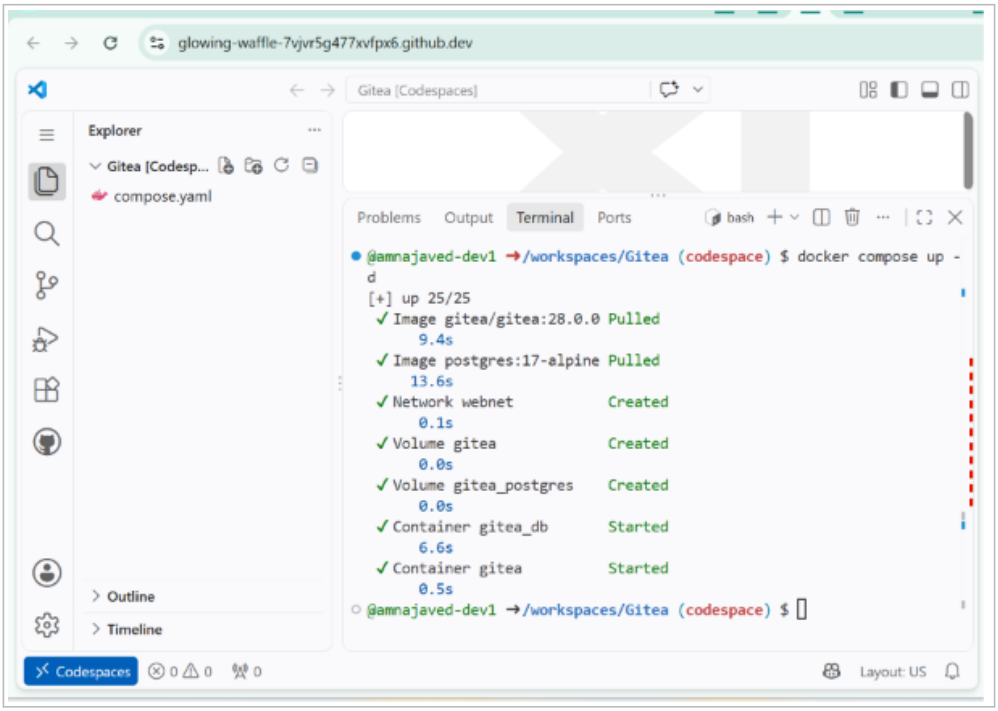
forked_gitea.png

2. Created and loaded the Codespace.



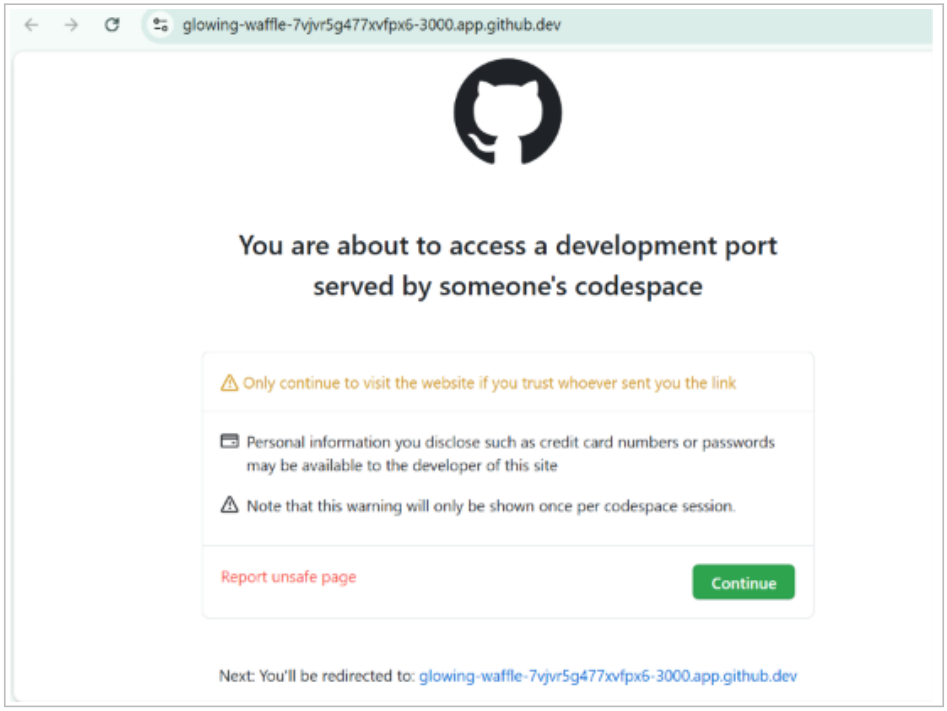
codespace_loading.png

3. Switched to the `codespace` branch and started containers with `docker compose up -d`.



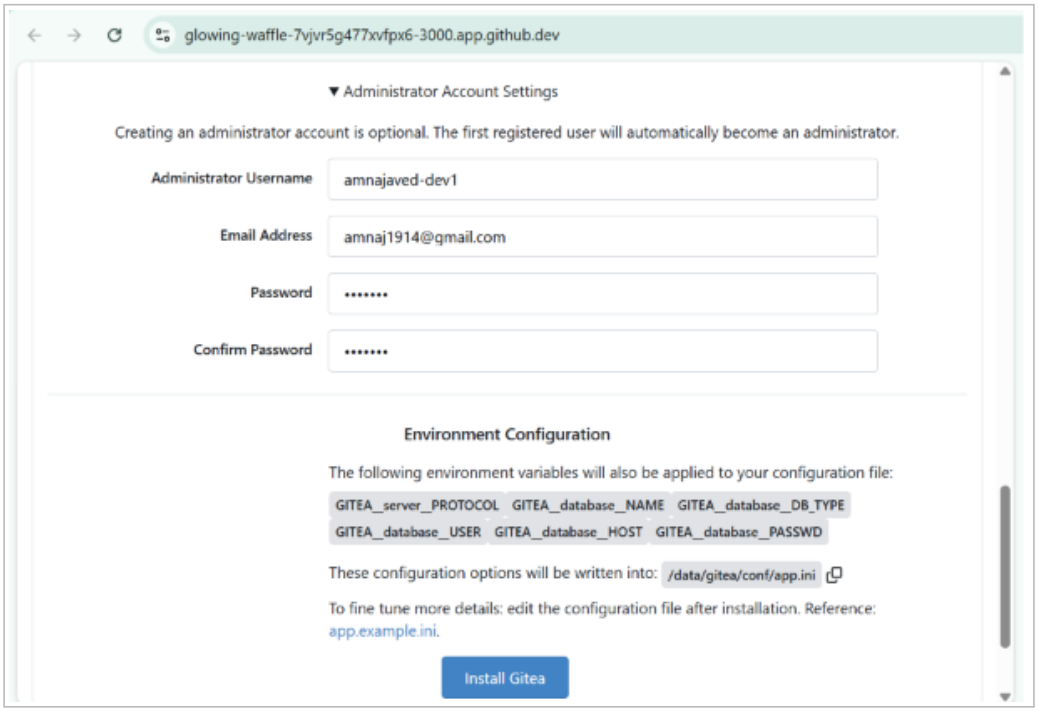
docker_up.png

4. Forwarded port 3000 to public visibility and opened the install interface.



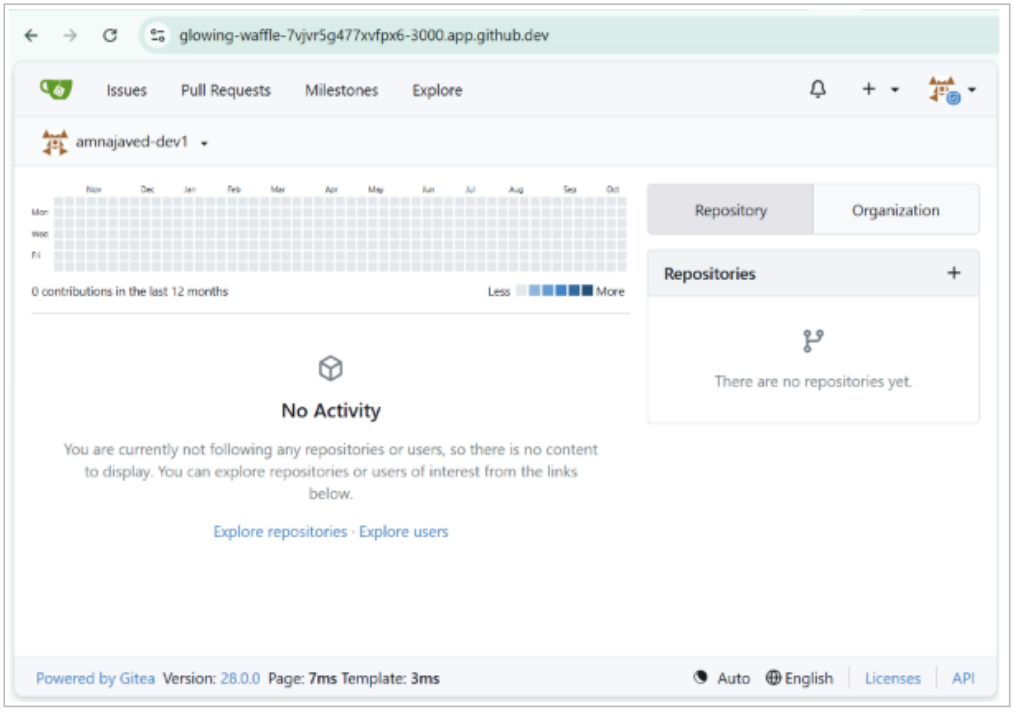
gitea_install_page.png

5. Configured the admin account setup.



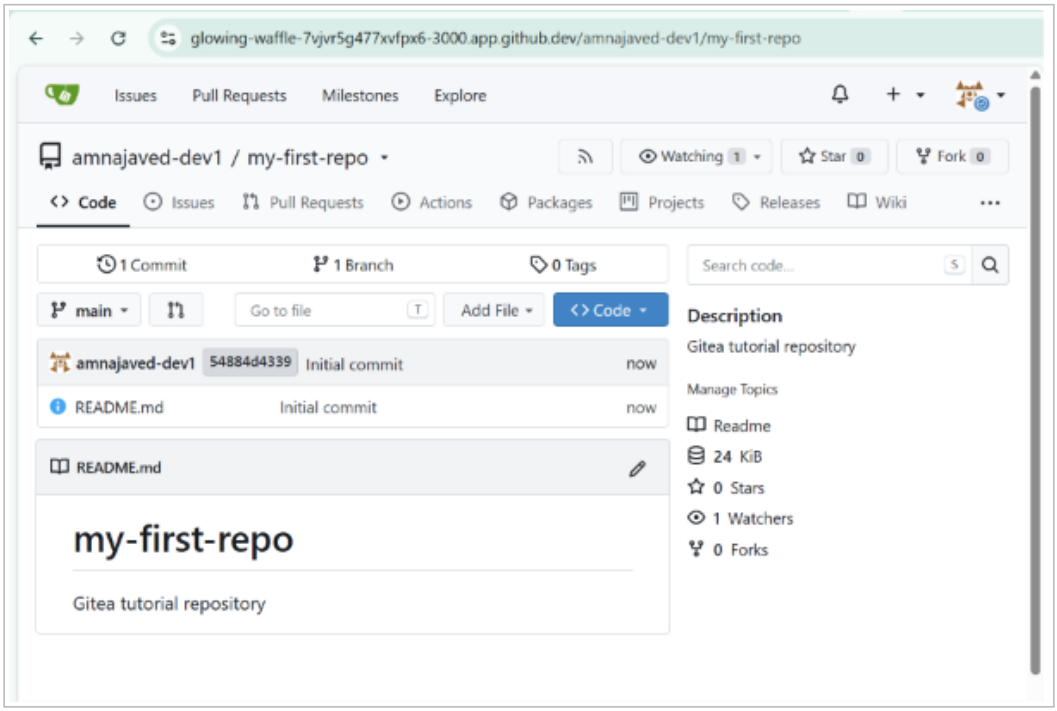
admin_setup.png

6. Logged in and verified the Gitea dashboard.



gitea_dashboard.png

7. Created a test repository inside Gitea (`my-first-repo`).



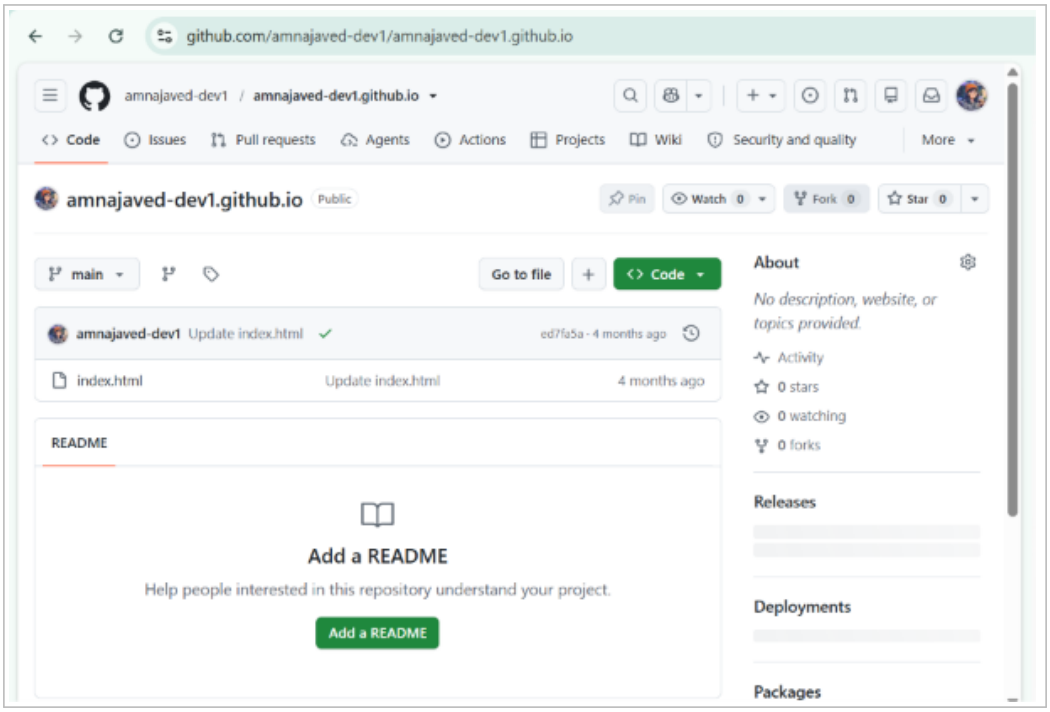
gitea_new_repo.png

Task 11: Creating a GitHub Pages Portfolio Site

Objective: Deploy a personal static developer portfolio using GitHub Pages.

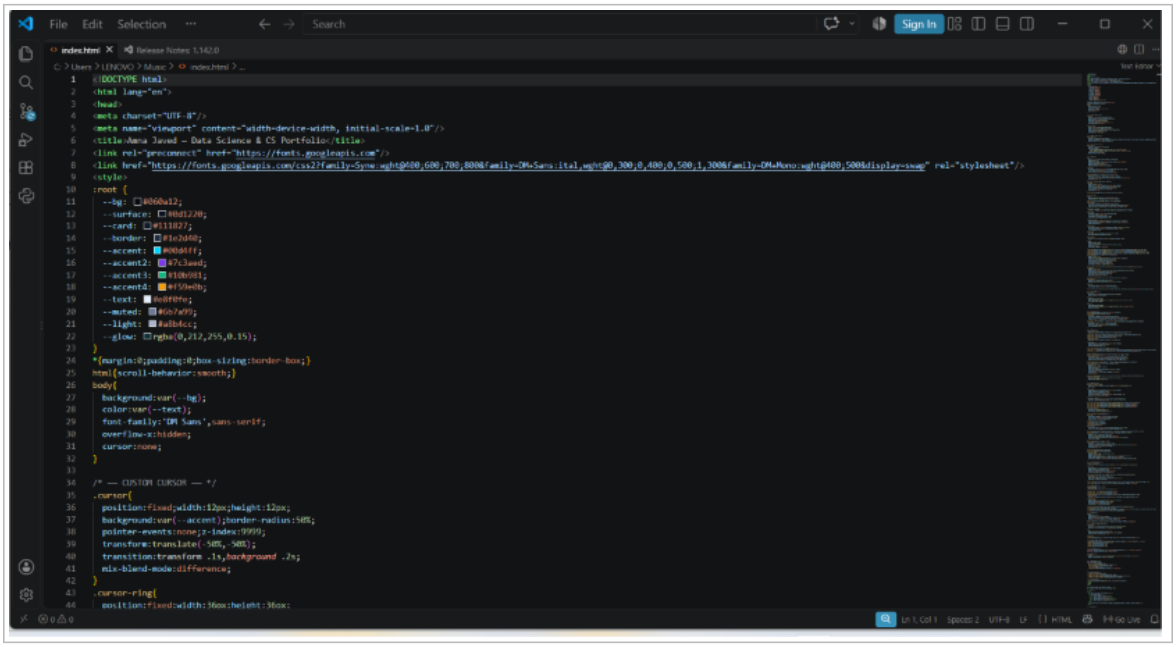
Steps Performed:

1. Created a public GitHub repository named `amnajaved-dev1.github.io`.



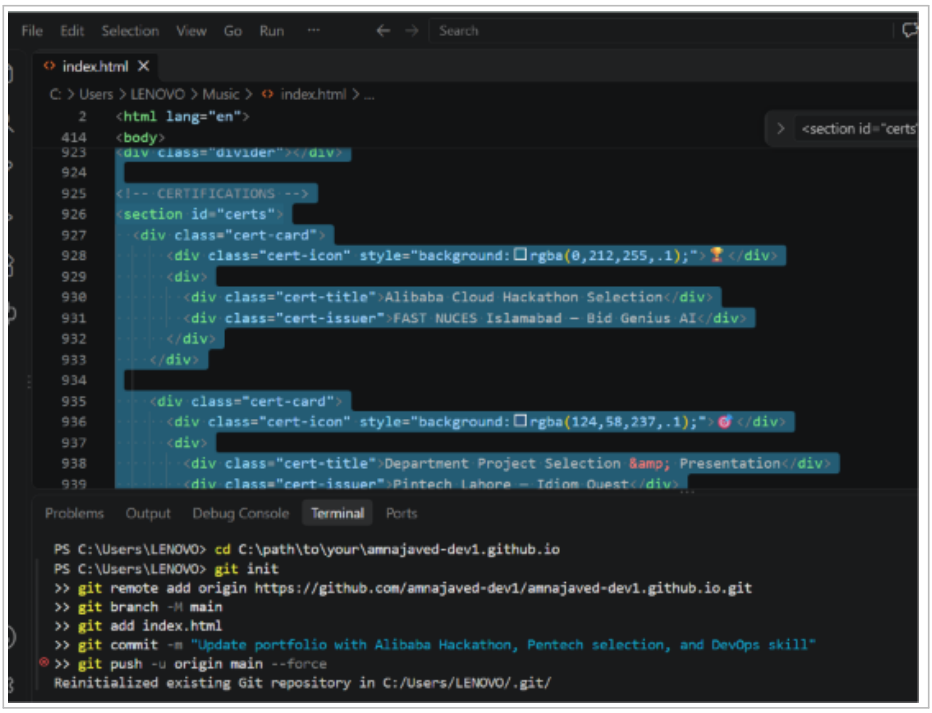
github_pages_repo.png

2. Prepared local portfolio files (`index.html`).



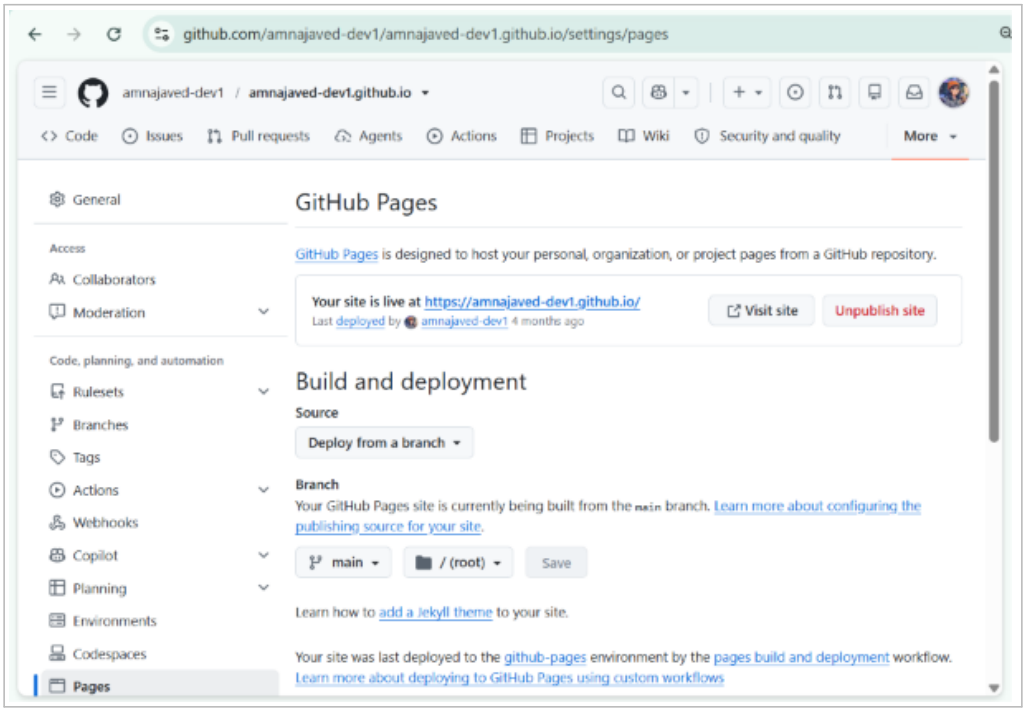
local_static_site.png

3. Initialized git and pushed the static files to `main`.



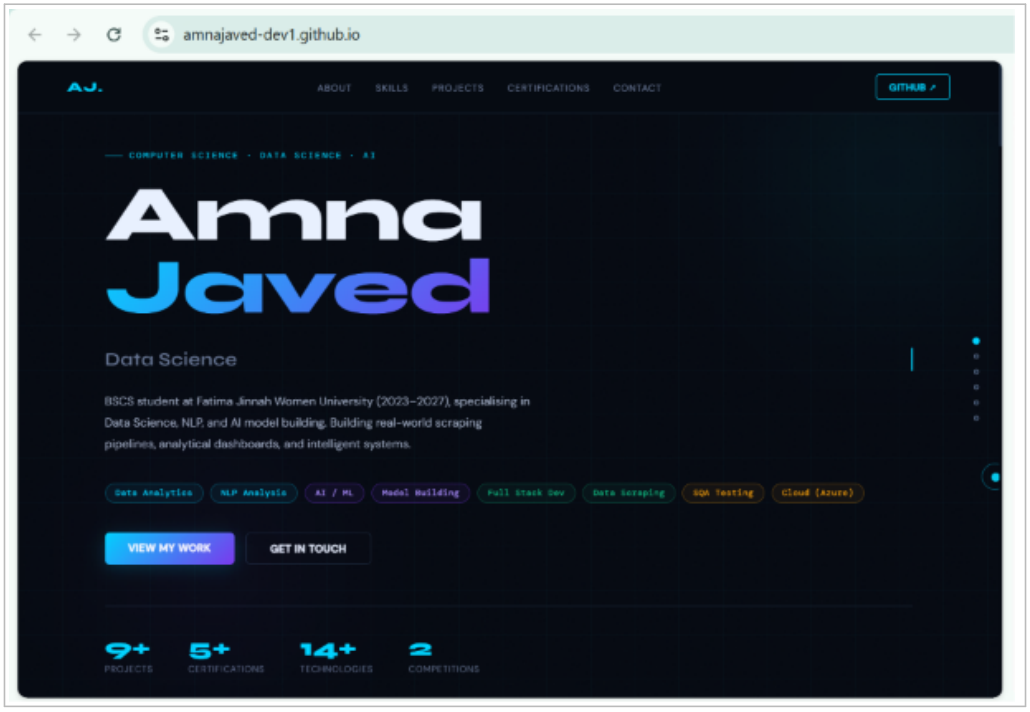
push_static_site.png

4. Confirmed deployment settings under repository Settings > Pages.



github_pages_settings.png

5. Verified the live portfolio site.

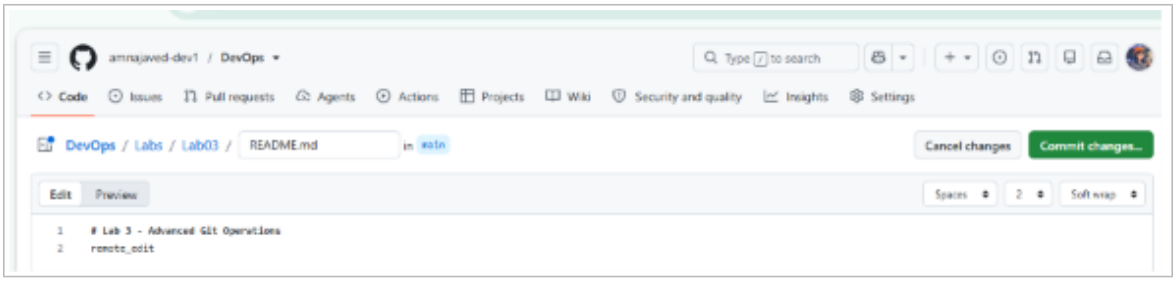


live_site.png

Exam Evaluation Questions

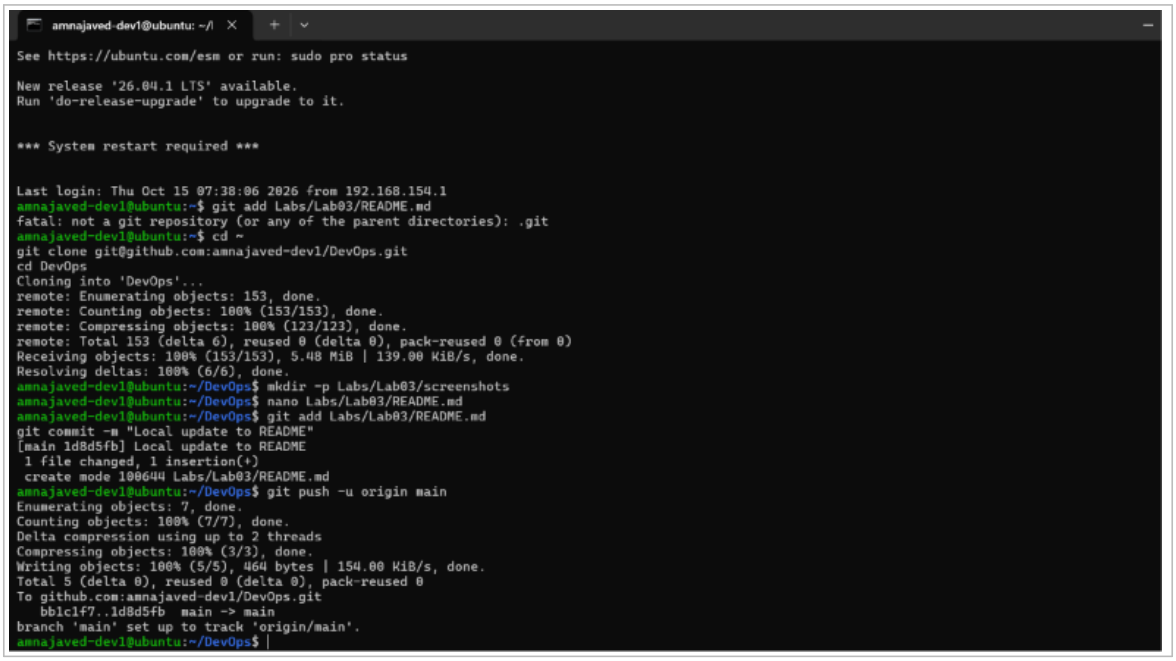
1. Local vs Remote Conflict Resolution

Step 1 (Remote Edit)



remote_edit.png

Step 2 (Local Edit)



local_commit.png

Step 3 (Push Error)



push_error.png

Step 4 (Merge Resolution & Push)

```
amnajaved-dev1@ubuntu:~/DevOps$ git pull --no-rebase origin main
remote: Enumerating objects: 9, done.
remote: Counting objects: 100% (9/9), done.
remote: Compressing objects: 100% (3/3), done.
remote: Total 5 (delta 1), reused 0 (delta 0), pack-reused 0 (from 0)
Unpacking objects: 100% (5/5), 1.06 KiB | 120.00 KiB/s, done.
From github.com:amnajaved-dev1/DevOps
 * branch            main       -> FETCH_HEAD
   1d8d5fb..015e935  main       -> origin/main
Updating 1d8d5fb..015e935
Fast-forward
 Labs/Lab03/README.md | 1 +
 1 file changed, 1 insertion(+)
```

merge_commit.png

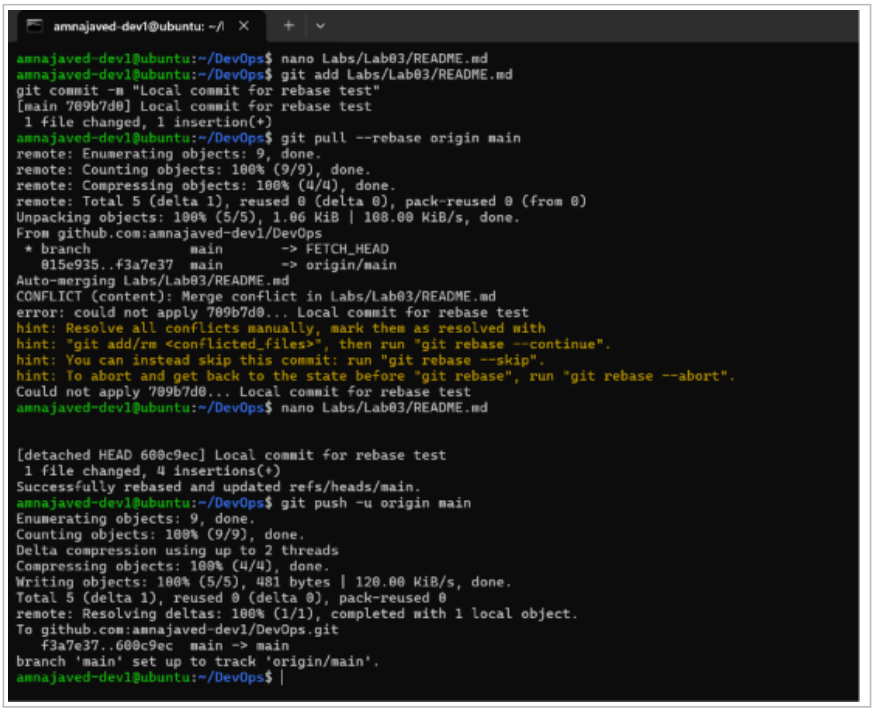
```
amnajaved-dev1@ubuntu:~/DevOps$ git push -u origin main
branch 'main' set up to track 'origin/main'.
Everything up-to-date
amnajaved-dev1@ubuntu:~/DevOps$
```

push_after_merge.png

Step 5 (Rebase Resolution & Push)

```
amnajaved-dev1@ubuntu:~/DevOps$ git push -u origin main
branch 'main' set up to track 'origin/main'.
Everything up-to-date
amnajaved-dev1@ubuntu:~/DevOps$ nano Labs/Lab03/README.md
amnajaved-dev1@ubuntu:~/DevOps$ git add Labs/Lab03/README.md
git commit -m "Local commit for rebase test"
[main 709b7d0] Local commit for rebase test
 1 file changed, 1 insertion(+)
amnajaved-dev1@ubuntu:~/DevOps$ git pull --rebase origin main
remote: Enumerating objects: 9, done.
remote: Counting objects: 100% (9/9), done.
remote: Compressing objects: 100% (4/4), done.
remote: Total 5 (delta 1), reused 0 (delta 0), pack-reused 0 (from 0)
Unpacking objects: 100% (5/5), 1.06 KiB | 108.00 KiB/s, done.
From github.com:amnajaved-dev1/DevOps
 * branch            main       -> FETCH_HEAD
   015e935..f3a7e37  main       -> origin/main
Auto-merging Labs/Lab03/README.md
CONFLICT (content): Merge conflict in Labs/Lab03/README.md
error: could not apply 709b7d0... Local commit for rebase test
hint: Resolve all conflicts manually, mark them as resolved with
hint: "git add/rm <conflicted_files>", then run "git rebase --continue".
hint: You can instead skip this commit: run "git rebase --skip".
hint: To abort and get back to the state before "git rebase", run "git rebase --abort".
Could not apply 709b7d0... Local commit for rebase test
amnajaved-dev1@ubuntu:~/DevOps$
```

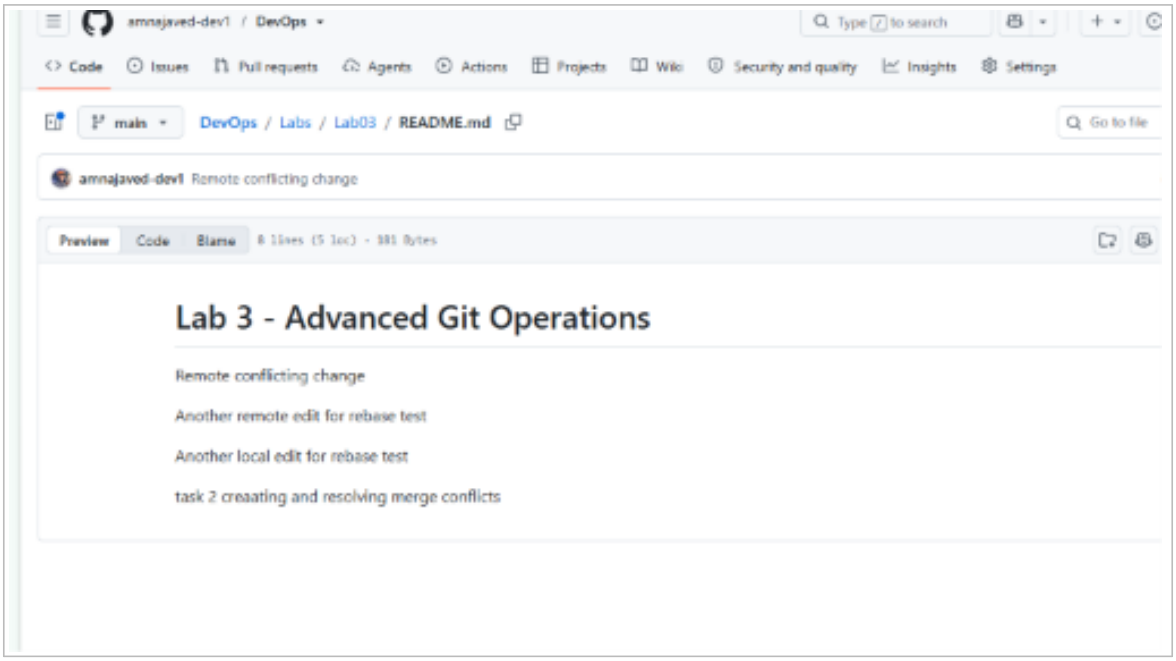
rebase_pull.png



push_after_rebase.png

2. Manual Merge Conflict Handling

Step 1 (Remote Conflict Edit)



remote_conflict_edit.png

Step 2 (Local Conflict Edit)

[Screenshot not found: local_conflict_edit.png]

local_conflict_edit.png

Step 3 (Conflict Push Error)

```
amnajaved-dev1@ubuntu:~/DevOps$ nano Labs/Lab03/README.md
amnajaved-dev1@ubuntu:~/DevOps$ git add Labs/Lab03/README.md
git commit -m "Local conflicting change"
[main 18a8f9e] Local conflicting change
 1 file changed, 1 insertion(+), 1 deletion(-)
amnajaved-dev1@ubuntu:~/DevOps$ git push -u origin main
To github.com:amnajaved-dev1/DevOps.git
 ! [rejected]        main -> main (fetch first)
error: failed to push some refs to 'github.com:amnajaved-dev1/DevOps.git'
hint: Updates were rejected because the remote contains work that you do not
hint: have locally. This is usually caused by another repository pushing to
hint: the same ref. If you want to integrate the remote changes, use
hint: 'git pull' before pushing again.
hint: See the 'Note about fast-forwards' in 'git push --help' for details.
amnajaved-dev1@ubuntu:~/DevOps$
```

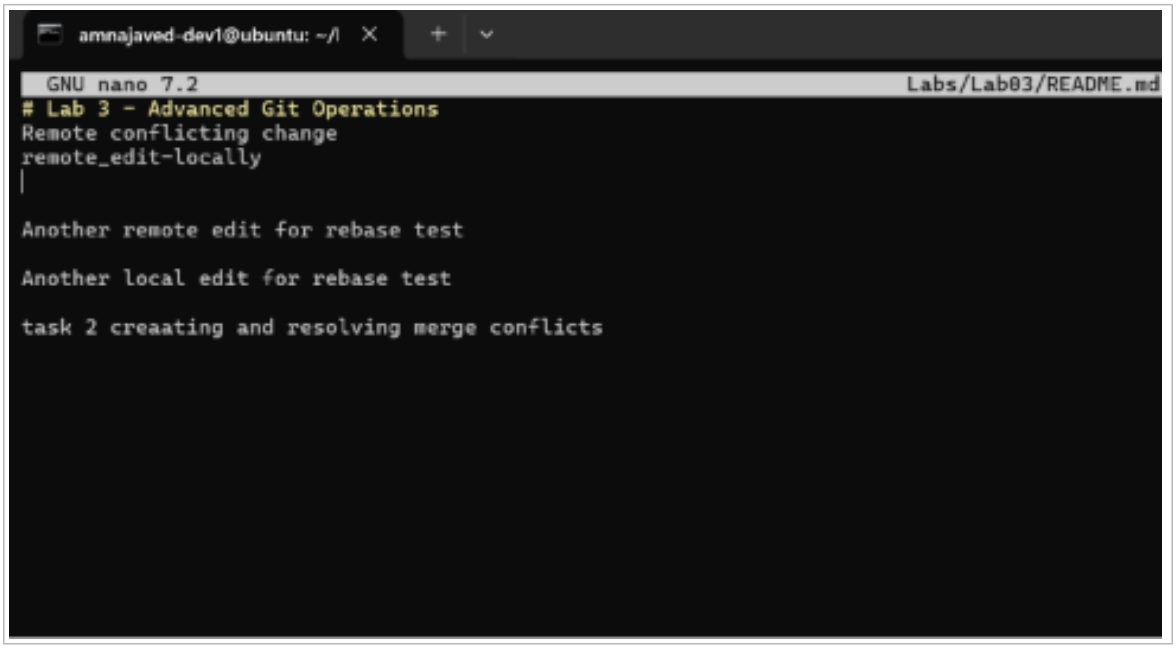
conflict_push_error.png

Step 4 (Rebase Conflict Trigger)

```
amnajaved-dev1@ubuntu:~/DevOps$ nano Labs/Lab03/README.md
amnajaved-dev1@ubuntu:~/DevOps$ git add Labs/Lab03/README.md
git commit -m "Local conflicting change"
[main 18a8f9e] Local conflicting change
 1 file changed, 1 insertion(+), 1 deletion(-)
amnajaved-dev1@ubuntu:~/DevOps$ git push -u origin main
To github.com:amnajaved-dev1/DevOps.git
 ! [rejected]        main -> main (fetch first)
error: failed to push some refs to 'github.com:amnajaved-dev1/DevOps.git'
hint: Updates were rejected because the remote contains work that you do not
hint: have locally. This is usually caused by another repository pushing to
hint: the same ref. If you want to integrate the remote changes, use
hint: 'git pull' before pushing again.
hint: See the 'Note about fast-forwards' in 'git push --help' for details.
amnajaved-dev1@ubuntu:~/DevOps$ git pull --rebase origin main
remote: Enumerating objects: 9, done.
remote: Counting objects: 100% (9/9), done.
remote: Compressing objects: 100% (4/4), done.
remote: Total 5 (delta 2), reused 0 (delta 0), pack-reused 0 (from 0)
Unpacking objects: 100% (5/5), 1.05 KiB | 35.00 KiB/s, done.
From github.com:amnajaved-dev1/DevOps
 * branch            main       -> FETCH_HEAD
   c5a1ff1..86aac14  main       -> origin/main
Auto-merging Labs/Lab03/README.md
CONFLICT (content): Merge conflict in Labs/Lab03/README.md
error: could not apply 18a8f9e... Local conflicting change
hint: Resolve all conflicts manually, mark them as resolved with
hint: "git add/rm <conflicted_files>", then run "git rebase --continue".
hint: You can instead skip this commit: run "git rebase --skip".
hint: To abort and get back to the state before "git rebase", run "git rebase --abort".
Could not apply 18a8f9e... Local conflicting change
amnajaved-dev1@ubuntu:~/DevOps$
```

conflict_message.png

Step 5 (Resolved File)

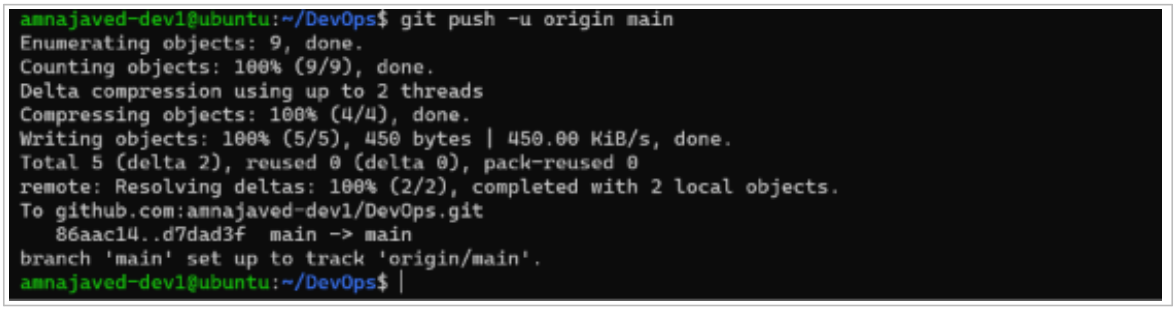


resolved_readme.png

Step 6 (Resolution Complete & Push)



rebase_continue.png



push_after_resolve.png

3. Managing Ignored and Tracked Files

Step 1 (Folder Created)

```
amnajaved-dev1@ubuntu:~/DevOps$ mkdir textfiles
echo "File A content" > textfiles/a.txt
echo "File B content" > textfiles/b.txt
echo "File C content" > textfiles/c.txt
amnajaved-dev1@ubuntu:~/DevOps$ git add .
git commit -m "Added textfiles directory with three files"
git push origin main
[main 4774fa5] Added textfiles directory with three files
 3 files changed, 3 insertions(+)
 create mode 100644 textfiles/a.txt
 create mode 100644 textfiles/b.txt
 create mode 100644 textfiles/c.txt
To github.com:amnajaved-dev1/DevOps.git
 ! [rejected]        main -> main (fetch first)
error: failed to push some refs to 'github.com:amnajaved-dev1/DevOps.git'
hint: Updates were rejected because the remote contains work that you do not
hint: have locally. This is usually caused by another repository pushing to
hint: the same ref. If you want to integrate the remote changes, use
hint: 'git pull' before pushing again.
hint: See the 'Note about fast-forwards' in 'git push --help' for details.
amnajaved-dev1@ubuntu:~/DevOps$ git pull --rebase origin main
remote: Enumerating objects: 9, done.
remote: Counting objects: 100% (9/9), done.
remote: Compressing objects: 100% (4/4), done.
remote: Total 5 (delta 2), reused 0 (delta 0), pack-reused 0 (from 0)
Unpacking objects: 100% (5/5), 1.02 KiB | 74.00 KiB/s, done.
From github.com:amnajaved-dev1/DevOps
 * branch            main       -> FETCH_HEAD
   d7dad3f..6728cf3  main       -> origin/main
Successfully rebased and updated refs/heads/main.
amnajaved-dev1@ubuntu:~/DevOps$ git push -u origin main
Enumerating objects: 7, done.
Counting objects: 100% (7/7), done.
Delta compression using up to 2 threads
Compressing objects: 100% (3/3), done.
Writing objects: 100% (6/6), 501 bytes | 250.00 KiB/s, done.
Total 6 (delta 0), reused 0 (delta 0), pack-reused 0
To github.com:amnajaved-dev1/DevOps.git
   6728cf3..f738e18  main -> main
branch 'main' set up to track 'origin/main'.
amnajaved-dev1@ubuntu:~/DevOps$
```

push_textfiles.png

Step 2 (Files Pushed)


```
amnajaved-dev1@ubuntu:~/DevOps$ mkdir textfiles
echo "File A content" > textfiles/a.txt
echo "File B content" > textfiles/b.txt
echo "File C content" > textfiles/c.txt
amnajaved-dev1@ubuntu:~/DevOps$ git add .
git commit -m "Added textfiles directory with three files"
git push origin main
[main 4774fa5] Added textfiles directory with three files
 3 files changed, 3 insertions(+)
 create mode 100644 textfiles/a.txt
 create mode 100644 textfiles/b.txt
 create mode 100644 textfiles/c.txt
To github.com:amnajaved-dev1/DevOps.git
 ! [rejected]        main -> main (fetch first)
error: failed to push some refs to 'github.com:amnajaved-dev1/DevOps.git'
hint: Updates were rejected because the remote contains work that you do not
hint: have locally. This is usually caused by another repository pushing to
hint: the same ref. If you want to integrate the remote changes, use
hint: 'git pull' before pushing again.
hint: See the 'Note about fast-forwards' in 'git push --help' for details.
amnajaved-dev1@ubuntu:~/DevOps$ git pull --rebase origin main
remote: Enumerating objects: 9, done.
remote: Counting objects: 100% (9/9), done.
remote: Compressing objects: 100% (4/4), done.
remote: Total 5 (delta 2), reused 0 (delta 0), pack-reused 0 (from 0)
Unpacking objects: 100% (5/5), 1.02 KiB | 74.00 KiB/s, done.
From github.com:amnajaved-dev1/DevOps
 * branch            main       -> FETCH_HEAD
   d7dad3f..6728cf3  main       -> origin/main
Successfully rebased and updated refs/heads/main.
amnajaved-dev1@ubuntu:~/DevOps$ git push -u origin main
Enumerating objects: 7, done.
Counting objects: 100% (7/7), done.
Delta compression using up to 2 threads
Compressing objects: 100% (3/3), done.
Writing objects: 100% (6/6), 501 bytes | 250.00 KiB/s, done.
Total 6 (delta 0), reused 0 (delta 0), pack-reused 0
To github.com:amnajaved-dev1/DevOps.git
   6728cf3..f738e18  main -> main
branch 'main' set up to track 'origin/main'.
amnajaved-dev1@ubuntu:~/DevOps$
```


push_textfiles.png

Step 3 (Gitignore Added)

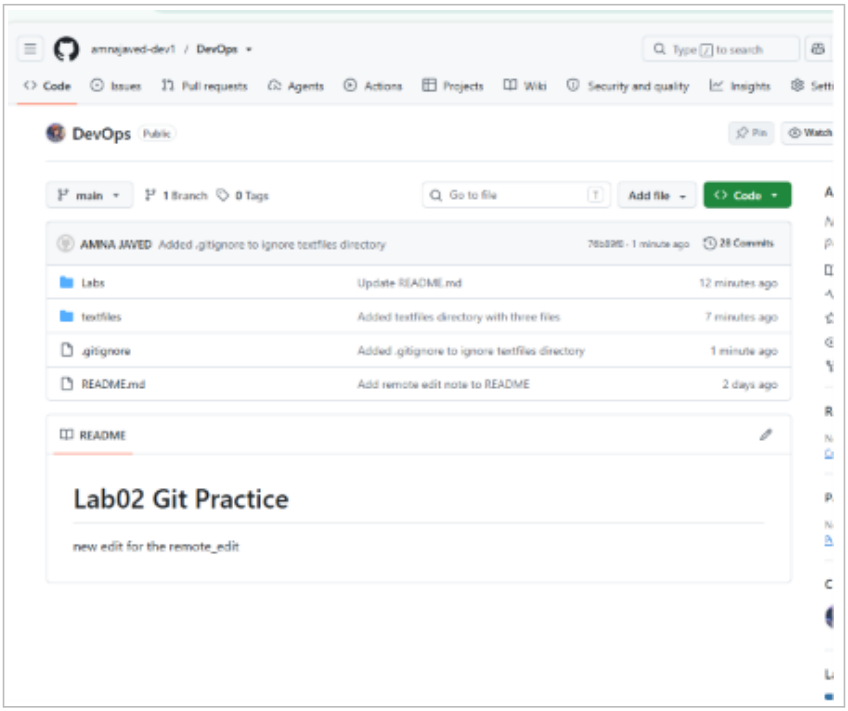
```
amnajaved-dev1@ubuntu:~/DevOps$ echo "textfiles/*" > .gitignore
amnajaved-dev1@ubuntu:~/DevOps$ git add .gitignore
git commit -m "Added .gitignore to ignore textfiles directory"
git push -u origin main
[main 76b89f8] Added .gitignore to ignore textfiles directory
 1 file changed, 1 insertion(+)
 create mode 100644 .gitignore
Enumerating objects: 4, done.
Counting objects: 100% (4/4), done.
Delta compression using up to 2 threads
Compressing objects: 100% (2/2), done.
Writing objects: 100% (3/3), 381 bytes | 127.00 KiB/s, done.
Total 3 (delta 0), reused 0 (delta 0), pack-reused 0
To github.com:amnajaved-dev1/DevOps.git
   f738e18..76b89f8  main -> main
branch 'main' set up to track 'origin/main'.
amnajaved-dev1@ubuntu:~/DevOps$
```

gitignore_push.png

Step 4 (Gitignore Pushed)


```
amnajaved-dev1@ubuntu:~/DevOps$ echo "textfiles/*" > .gitignore
amnajaved-dev1@ubuntu:~/DevOps$ git add .gitignore
git commit -m "Added .gitignore to ignore textfiles directory"
git push -u origin main
[main 76b89f8] Added .gitignore to ignore textfiles directory
 1 file changed, 1 insertion(+)
 create mode 100644 .gitignore
Enumerating objects: 4, done.
Counting objects: 100% (4/4), done.
Delta compression using up to 2 threads
Compressing objects: 100% (2/2), done.
Writing objects: 100% (3/3), 381 bytes | 127.00 KiB/s, done.
Total 3 (delta 0), reused 0 (delta 0), pack-reused 0
To github.com:amnajaved-dev1/DevOps.git
   f738e18..76b89f8  main -> main
branch 'main' set up to track 'origin/main'.
amnajaved-dev1@ubuntu:~/DevOps$
```

gitignore_push.png



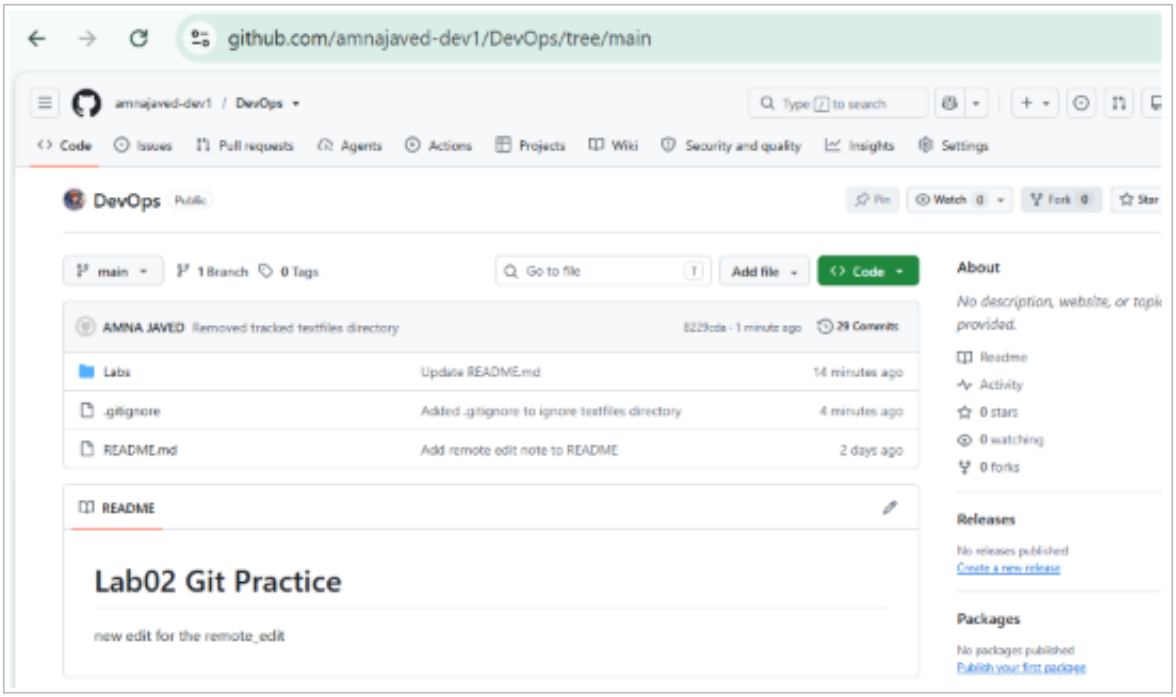
repo_still_has_textfiles.png

Step 5 (Folder Untracked)

```
amnajaved-dev1@ubuntu:~/DevOps$ git rm -r --cached textfiles
rm 'textfiles/a.txt'
rm 'textfiles/b.txt'
rm 'textfiles/c.txt'
amnajaved-dev1@ubuntu:~/DevOps$ git add .
git commit -m "Removed tracked textfiles directory"
git push -u origin main
[main 8229cda] Removed tracked textfiles directory
 3 files changed, 3 deletions(-)
 delete mode 100644 textfiles/a.txt
 delete mode 100644 textfiles/b.txt
 delete mode 100644 textfiles/c.txt
Enumerating objects: 3, done.
Counting objects: 100% (3/3), done.
Delta compression using up to 2 threads
Compressing objects: 100% (2/2), done.
Writing objects: 100% (2/2), 246 bytes | 246.00 KiB/s, done.
Total 2 (delta 1), reused 0 (delta 0), pack-reused 0
remote: Resolving deltas: 100% (1/1), completed with 1 local object.
To github.com:amnajaved-dev1/DevOps.git
   76b89f8..8229cda  main -> main
branch 'main' set up to track 'origin/main'.
amnajaved-dev1@ubuntu:~/DevOps$
```

rm_cached_push.png

Step 6 (Folder Removed from GitHub)



repo_textfiles_removed.png

4. Commit History Manipulation and Recovery

Step 1 (First Commit)

```
amnajaved-dev1@ubuntu:~/DevOps$ echo "First test line" >> Labs/Lab03/README.md
git add .
git commit -m "Added first test line"
[main 9a9b225] Added first test line
 1 file changed, 1 insertion(+)
amnajaved-dev1@ubuntu:~/DevOps$
```

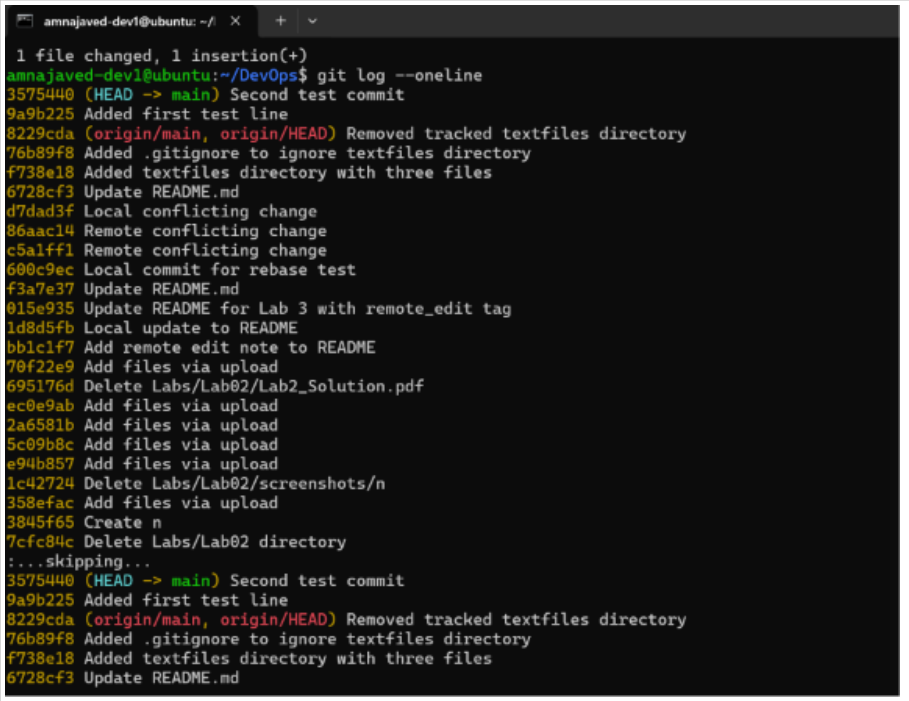
first_commit.png

Step 2 (Second Commit)

```
amnajaved-dev1@ubuntu:~/DevOps$ echo "Second test line" >> Labs/Lab03/README.md
git add .
git commit -m "Second test commit"
[main 3575440] Second test commit
 1 file changed, 1 insertion(+)
amnajaved-dev1@ubuntu:~/DevOps$
```

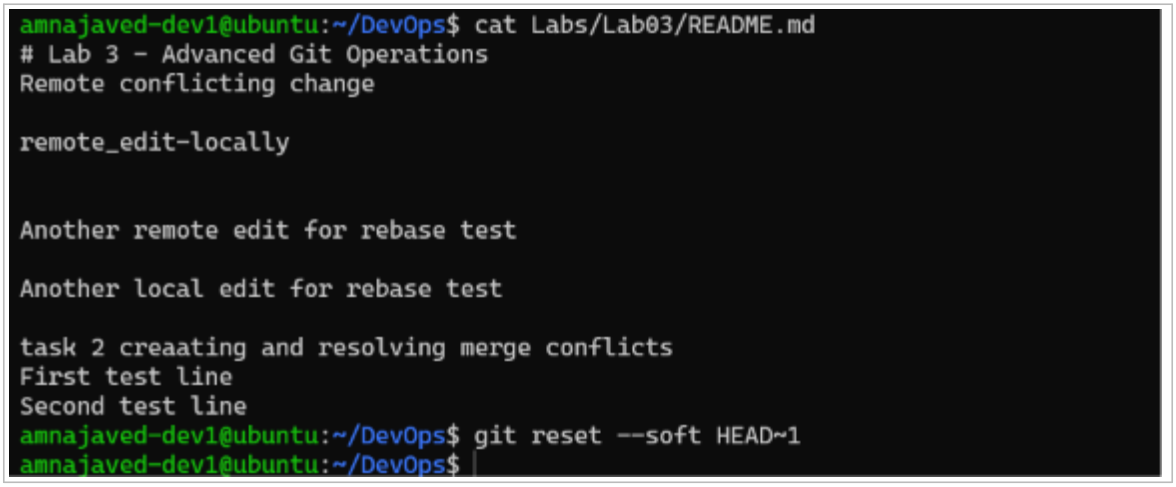
second_commit.png

Step 3 (Commit History)

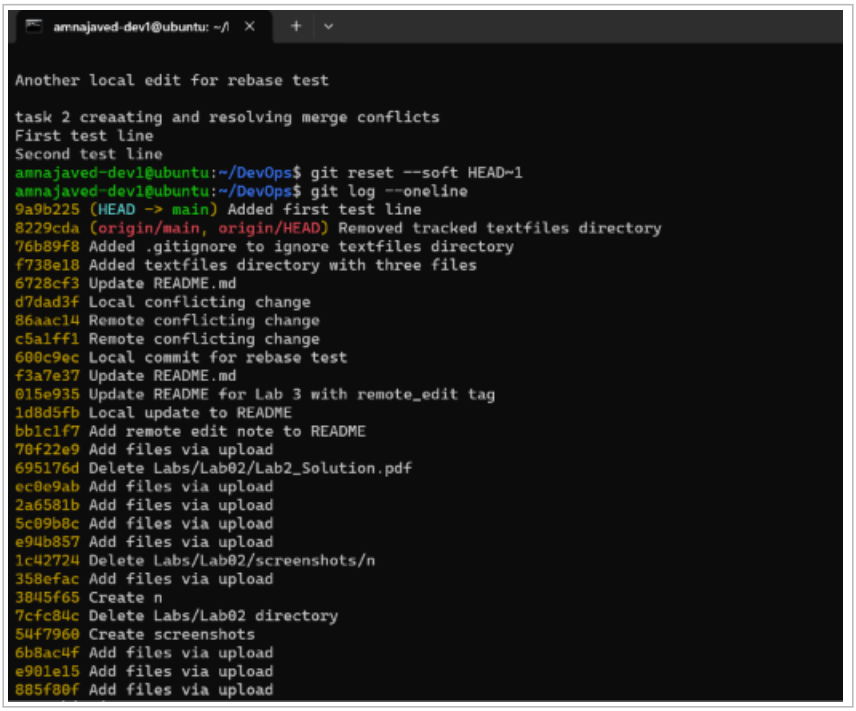


log_before_reset.png

Step 4 (Soft Reset)



soft_reset.png



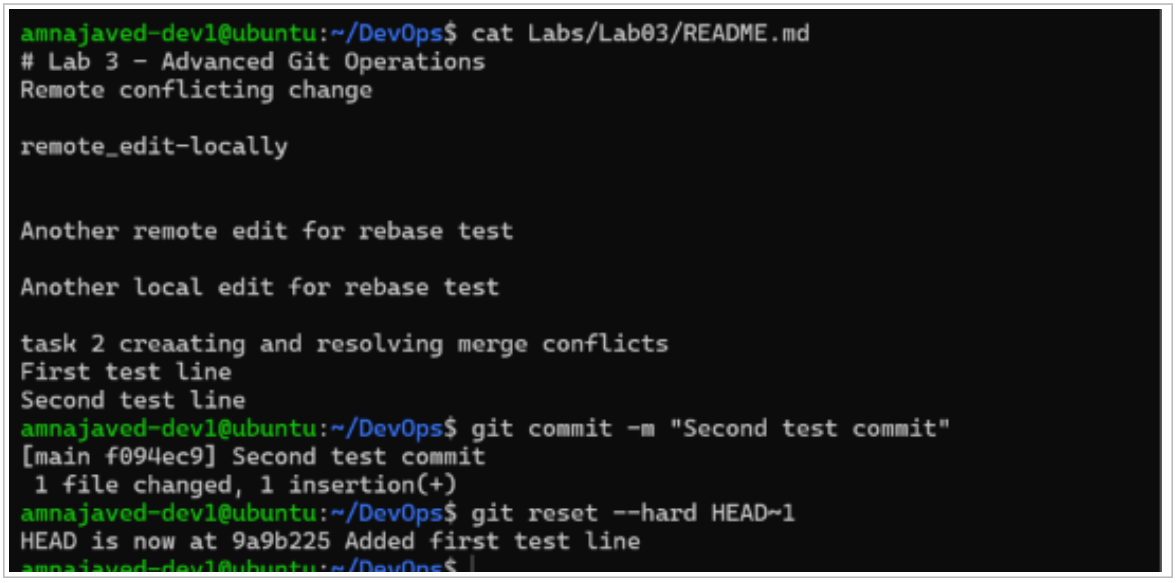
log_after_soft_reset.png

Step 5 (Third/Re-commit)

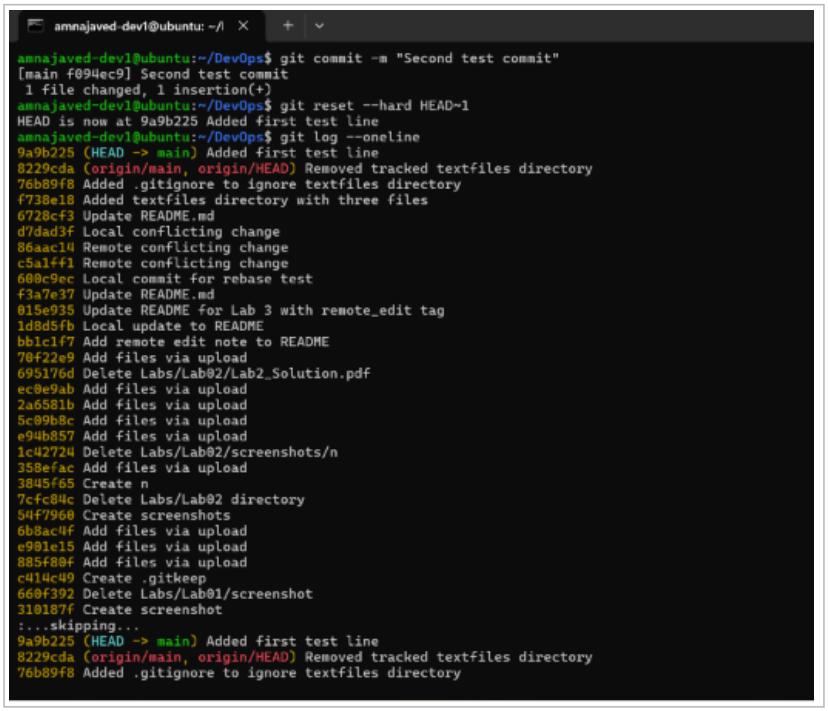
[Screenshot not found: third_commit.png]

third_commit.png

Step 6 (Hard Reset)



hard_reset.png



log_after_hard_reset.png

Repository Link: https://github.com/amnajaved-dev1/DevOps